

Supporting Information

$\Delta(\text{ICOB}I_{\text{antibonding}})$: A Fast Bond-Order-Based Descriptor for the Kinetic Viability of Metastable Crystalline Compounds

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S1 Dataset Composition and Selection Criteria

Candidate structures were drawn from the Materials Project database by querying unary and binary chemical systems, restricted to non-magnetic structures (absolute total magnetization $\leq 0.01 \mu_{\text{B}}$ per formula unit, computed directly from the stored magnetic moments) and excluding *f*-block elements. Three complementary selection tracks were combined into the final 597-structure dataset:

- **Track 1** (186 compounds): ≤ 20 atoms/cell, balanced across stability classes (60 per family), with a cap of two entries per chemical system to maximize elemental diversity over redundant polymorphs.
- **Track 2** (163 compounds/elements): ≤ 40 atoms/cell, targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N, O, F, P, S, and Cl, chosen to build genuine same-composition polymorph groups and probe descriptor behavior far from equilibrium.
- **Track 3** (45 binary compounds): an adversarial robustness set pairing an experimentally known compound sitting farthest above the convex hull of its chemical system with one to three theoretical polymorphs of the same formula lying even farther above the hull.

Table S1 summarizes the dataset by element count and Table S2 by stability class. One trivial-cell reference element yielded no usable LOBSTER step and is excluded from every ICOHP/ICOBI-derived statistic (Table S1, footnote).

Table S1: Dataset composition by element count.

Composition	<i>n</i>
Single-element references	69
Multi-element compounds	522
Additional structures without compound metadata ^a	6
Total	597

^a isolated gas dimers (H₂, N₂, O₂, F₂, Cl₂) and a CaO sphalerite-type reference, used only for internal quality-control checks (Section S6), outside the main dataset.

Table S2: Stability class, by E_{hull} and experimental/theoretical character.

Class	<i>n</i>	E_{hull} range (eV/atom)
Stable ($E_{\text{hull}} = 0$)	160	0.0000
Experimental metastable	183	0.0000347 – 2.1080
Theoretical metastable	242	0.0000843 – 3.1688
Unknown ^a	12	—
Total	597	

^a entries lacking a usable E_{hull} value or predating the `theoretical` flag used to build later batches of this dataset.

S2 VASP Calculation Parameters

Table S3 lists the common INCAR parameters applied identically to every structure in the dataset (Section 2.3 of the main text). Table S4 lists the PAW-PBE potential used for each element (pymatgen’s MPRelaxSet library, with plain W substituted for W_sv/W_pv after the latter was found to be missing its LEXCH field in the local potential mirror; Section S6).

Table S3: Common VASP INCAR parameters.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (−1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMEAR	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (−1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

Table S4: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv (§S6)).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Bi	Bi (08Apr2002)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)

Element	PAW-PBE potential
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)
Zr	Zr_sv (04Jan2005)

S3 k -Point Grid and NBANDS per Compound

Table S5 lists, for the 186 compounds of Track 1 (Section S1), the automatic k -point grid (`pymatgen` automatic density, `kppv01=100`; G/M = Γ - or Monkhorst–Pack-centered) and the number of bands (NBANDS, derived from the number of local orbitals required by the LOBSTER basis for that compound). The same generation rule was applied to every other track of the dataset; the complete machine-readable listing for all 597 structures is available in the project’s data repository (Data Availability, main text).

Table S5: k -point grid (`pymatgen` automatic density, `kppv01=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k -point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154

Table S5 (continued)

Formula	mp-id	k-point grid	NBANDS
ScAg ₂	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO ₂	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta ₅ Ge ₃	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb ₂	mp-11697	8 5 4 (G)	34
Te ₂ Pd	mp-782	7 7 5 (G)	17
Te ₂ Ru	mp-267	7 5 4 (G)	34
TeO ₂	mp-8377	6 5 3 (G)	48
Ti ₃ In	mp-866046	5 5 6 (G)	72
Ti ₃ Pt	mp-2339	5 5 5 (G)	72
TiBe ₁₂	mp-1104067	7 5 5 (G)	69
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi ₂	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi ₂	mp-11531	12 8 7 (G)	27
WO ₂	mp-1524394	6 5 5 (G)	68
Zr ₂ Au	mp-2348	9 9 4 (G)	29
Zr ₅ Pb ₃	mp-681992	3 3 5 (G)	154
ZrMo ₂	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI ₂	mp-567259	7 7 4 (G)	17
MgCl ₂	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li ₃ Hg	mp-1646	7 7 7 (G)	24
Ta ₇ Co ₆	mp-1104548	6 6 3 (G)	117
CaGa ₄	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH ₂	mp-23711	6 5 5 (G)	24
Sn ₇ Ir ₅	mp-20466	4 4 4 (M)	108
NaN ₃	mp-1066400	7 7 7 (G)	16
ClF ₃	mp-556767	6 4 3 (G)	64
SiS ₂	mp-1095294	5 4 3 (G)	48
Sr ₅ Sn ₃	mp-17720	3 3 3 (G)	104
Te ₂ Ir	mp-1551	7 5 2 (G)	51
Li ₁₃ Sn ₅	mp-30769	6 6 1 (G)	110
H ₃ N	mp-643432	8 5 5 (G)	28
Cs ₂ Se	mp-1011696	5 3 2 (G)	56
In ₃ Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb ₂ Hg ₇	mp-31474	4 4 4 (G)	73
PbAu ₂	mp-19871	5 5 5 (G)	54

Table S5 (continued)

Formula	mp-id	k-point grid	NBANDS
BeCu	mp-2323	10 10 10 (M)	100
SiO ₂	mp-546794	6 6 6 (M)	24
IF ₇	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al ₃ Os ₂	mp-16521	9 9 3 (G)	30
Nb ₂ O ₅	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd ₃	mp-7753	7 7 7 (G)	31
AsPd ₂	mp-1080831	8 4 4 (G)	66
GeSe ₂	mp-10074	5 5 5 (G)	34
Re ₃ Ge ₇	mp-29141	9 6 1 (G)	180
Cu ₃ Ge	mp-19724	6 6 5 (G)	72
Ti ₃ Ge	mp-1189044	4 4 4 (M)	144
ZrGa ₃	mp-30685	7 7 5 (G)	37
SrAl ₂	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al ₁₂ W	mp-11227	4 4 4 (M)	57
CsH	mp-632319	7 7 7 (G)	6
CaGa ₂	mp-992	7 7 6 (G)	23
NiHg ₄	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn ₅	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te ₂ Au	mp-567525	7 7 5 (G)	17
Be ₃ N ₂	mp-6977	10 10 3 (G)	46
Y ₂ O ₃	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe ₂	mp-1102002	4 4 4 (M)	68
H ₂ O	mp-2739191	7 7 4 (G)	24
CdO ₂	mp-2310	5 5 5 (G)	68
Mo ₃ Se ₄	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be ₁₂ Pd	mp-12666	6 6 6 (M)	69
Sr ₂ As	mp-15697	6 6 3 (G)	28
AsRh ₂	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K ₂ S	mp-559278	5 5 4 (G)	28
Ge ₄ Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li ₃ Ag	mp-976408	7 7 7 (G)	24
Y ₃ Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr ₃ As ₄	mp-678007	3 3 4 (G)	62
Ga ₂ Se ₃	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl ₂	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13

Table S5 (continued)

Formula	mp-id	k-point grid	NBANDS
TiW	mp-1216621	10 10 6 (M)	18
SrTi ₃	mp-1206636	5 5 5 (G)	32
IrCl ₃	mp-729507	3 3 6 (G)	84
Ta ₄ C ₃	mp-1218000	6 6 6 (M)	48
Sc ₂ O ₃	mp-755313	6 5 6 (G)	64
SnBr ₂	mp-978985	4 4 6 (M)	34
Sn ₇ Pd ₅	mp-1219006	5 5 3 (G)	108
Tl ₄ Sn	mp-1216560	5 5 5 (G)	45
LiTi ₃	mp-1185253	6 6 6 (M)	32
PtCl ₂	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K ₅ As ₄	mp-684799	4 4 3 (G)	41
KSn ₃	mp-1185151	4 4 5 (G)	64
NbS ₂	mp-774873	10 6 2 (M)	34
YAg ₃	mp-1187811	6 6 6 (G)	37
B ₃ Ir ₂	mp-1228647	9 5 4 (G)	60
TiMo ₂	mp-1216675	5 6 14 (G)	27
Cr ₂ C	mp-1226378	10 10 7 (G)	22
ZrO ₂	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN ₂	mp-1097011	5 5 5 (G)	24
CoSe ₂	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo ₃	mp-1095057	7 6 6 (G)	72
Tl ₃ Ga	mp-1187539	5 5 5 (G)	36
CaGe ₂	mp-643542	7 7 6 (G)	23
NbRh ₂	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF ₂	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na ₂ Cl	mp-990084	6 6 6 (M)	12
ZnPb ₃	mp-1187949	5 5 5 (G)	36
Sr ₃ Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al ₄ Si	mp-1228120	6 6 6 (M)	20
B ₆ Pb	mp-1096825	6 6 6 (M)	33
ScTc ₃	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd ₅	mp-1228665	6 6 6 (M)	49
Zr ₃ Be	mp-1188016	6 6 6 (M)	35
Na ₃ Cl	mp-1064484	8 8 3 (G)	16
SiAu ₃	mp-1186998	7 7 7 (G)	31
InSe ₂	mp-1232036	7 7 3 (G)	34
TiSi ₂	mp-570991	8 8 8 (M)	17
BeAu ₃	mp-983539	7 7 7 (G)	32
Zr ₃ Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg ₁₅ B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

Table S5 (continued)

Formula	mp-id	k-point grid	NBANDS
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S4 Elemental Reference Structures

Table S6 lists the 62 elemental reference structures used as the products side of every case-1 (decomposition-into-elements) reaction, together with their raw and near-frontier antibonding ICOHP values (main text, Section 2.6).

Table S6: The 62 elemental reference structures used as the products side of every case-1 (decomposition-to-elements) reaction (SG = space group). $\Delta = E_{\text{campaign}} - E_{\text{MP}}$ per atom: the large offsets (up to several tens of eV/atom for heavy metals) reflect a PAW pseudopotential valence-electron convention difference (semi-core electrons explicitly included, `_d/_pv/_sv` variants), NOT a DFT-pipeline error: absolute total energies are only meaningful within one consistent pseudopotential set. N has no MP entry (hand-built dimer box).

Elem.	mp-id	SG	E_{MP} (eV)	E_{camp} (eV)	Δ (meV)	ICOHP	ICOHP antib.
Ag	mp-124	Fm-3m	-21.35	-2.71	18640	-1.250	0.000
Al	mp-134	Fm-3m	-6.73	-3.75	2979	-1.419	0.000
As	mp-158	Cmce	-14.71	-4.65	10060	-3.968	0.010
Au	mp-81	Fm-3m	-50.58	-3.23	47357	-1.338	0.000
B	mp-160	R-3m	-7.35	-6.71	641	-4.796	0.000
Ba	mp-122	Im-3m	-1.92	-1.90	19	-0.012	8.240
Be	mp-87	P6 ₃ /mmc	-3.74	-3.77	-27	-1.898	0.000
Br	mp-23154	Cmce	-1.64	-1.62	12	-3.536	0.426
C	mp-48	P6 ₃ /mmc	-9.94	-9.23	719	-11.539	0.000
Ca	mp-21	Im-3m	-7.03	-1.91	5123	-0.030	3.221
Cd	mp-94	P6 ₃ /mmc	-20.10	-0.73	19372	-1.612	0.000
Cl	mp-22848	Cmce	-1.85	-1.83	17	-5.573	0.456
Co	mp-102	Fm-3m	-13.23	-7.02	6211	-1.915	0.019
Cr	mp-90	Im-3m	-15.07	-9.38	5698	-0.624	0.011
Cs	mp-1	Im-3m	-25.09	-0.85	24238	-0.123	0.170
Cu	mp-30	Fm-3m	-10.85	-3.73	7112	-0.249	0.164
F	mp-561203	C2/c	-1.91	-1.80	109	-6.954	0.457
Fe	mp-13	Im-3m	-8.47	-8.26	206	-0.537	0.092
Ga	mp-142	Cmce	-11.44	-2.90	8547	-2.638	0.000
Ge	mp-32	Fd-3m	-13.87	-4.50	9368	-4.645	0.002
H	mp-730101	P2 ₁₂ ₁₂ ₁	-3.46	-3.39	67	-5.122	0.000
Hf	mp-103	P6 ₃ /mmc	-45.15	-9.93	35223	-1.161	0.000
Hg	mp-10861	P6 ₃ /mmm	-49.36	-0.18	49187	-0.965	0.000
I	mp-23153	Cmce	-1.52	-1.51	15	-0.150	0.498
In	mp-1055994	I4 ₁ /mmm	-22.57	-2.54	20035	-1.156	0.000
Ir	mp-101	Fm-3m	-52.36	-8.86	43495	-2.413	0.032
K	mp-10157	Fm-3m	-6.01	-1.04	4974	-0.081	0.092
Li	mp-1018134	R-3m	-2.39	-1.91	481	-0.693	0.000
Mg	mp-153	P6 ₃ /mmc	-4.17	-1.44	2733	0.000	—
Mn	mp-35	I-43m	-14.68	-8.84	5841	-0.608	0.041

Elem.	mp-id	SG	E_{MP} (eV)	E_{camp} (eV)	Δ (meV)	ICOHP	ICOHP antib.
Mo	mp-129	Im-3m	-26.07	-10.90	15174	-1.106	0.014
N	—	—	—	-8.33	—	—	—
Na	mp-10172	P6 ₃ /mmc	-3.56	-1.31	2253	-0.084	0.117
Nb	mp-75	Im-3m	-24.73	-10.08	14650	-1.144	0.000
Ni	mp-23	Fm-3m	-11.98	-5.48	6498	-0.420	0.380
O	mp-1524462	C2/m	-5.97	-4.86	1111	-17.796	0.127
Os	mp-49	P6 ₃ /mmc	-52.99	-11.23	41760	-1.158	0.054
P	mp-568348	P2/c	-9.00	-5.40	3598	-5.076	0.021
Pb	mp-20483	Fm-3m	-56.29	-3.58	52709	-0.852	0.028
Pd	mp-2	Fm-3m	-22.96	-5.20	17760	-1.216	0.126
Pt	mp-126	Fm-3m	-51.49	-6.10	45389	-1.809	0.093
Rb	mp-70	Im-3m	-0.97	-0.94	29	-0.118	0.131
Re	mp-8	P6 ₃ /mmc	-52.41	-12.39	40018	-1.380	0.021
Rh	mp-74	Fm-3m	-24.14	-7.26	16879	-0.708	0.185
Ru	mp-33	P6 ₃ /mmc	-25.42	-9.23	16196	-0.948	0.087
S	mp-77	Fddd	-8.03	-4.12	3915	-5.925	0.088
Sb	mp-104	R-3m	-25.71	-4.15	21561	-1.720	0.043
Sc	mp-67	P6 ₃ /mmc	-11.40	-6.24	5157	-0.660	0.000
Se	mp-14	P3 ₁ 21	-14.31	-3.50	10811	-4.150	0.083
Si	mp-149	Fd-3m	-8.77	-5.42	3352	-4.542	0.000
Sn	mp-623511	I4/mmm	-24.68	-3.79	20895	-0.880	0.023
Sr	mp-139	P6 ₃ /mmc	-15.15	-1.63	13515	-0.017	4.905
Ta	mp-50	Im-3m	-48.58	-11.82	36766	-1.279	0.000
Tc	mp-113	P6 ₃ /mmc	-25.96	-10.35	15615	-1.206	0.024
Te	mp-19	P3 ₁ 21	-25.42	-3.14	22279	-1.123	0.120
Ti	mp-46	P6 ₃ /mmc	-12.92	-7.80	5122	-0.754	0.000
Tl	mp-82	P6 ₃ /mmc	-53.18	-2.21	50972	-1.046	0.000
V	mp-146	Im-3m	-14.09	-8.96	5128	-0.878	0.000
W	mp-91	Im-3m	-51.37	-13.00	38368	-2.560	0.000
Y	mp-112	P6 ₃ /mmc	-20.31	-6.42	13896	-0.868	0.000
Zn	mp-79	P6 ₃ /mmc	-8.91	-1.09	7823	-1.889	0.000
Zr	mp-131	P6 ₃ /mmc	-22.75	-8.52	14227	-1.317	0.000

S5 Bonding-Type Classification: Full Counts

Table S7 reports, for the two independent bonding classifications discussed in the main text (Section 3.6), the full cross-tabulation over all 588 compounds and elements with usable ICOHP/ICOBI data.

The mixed (Zintl-type) category identifies compounds where a *homoatomic* pair above the covalent ICOBI threshold (e.g. N–N in an azide, P–P in a polyphosphide, S–S in a polysulfide) coexists with a markedly weaker *heteroatomic* pair linking a distinct cation, the structural signature of Zintl–Klemm chemistry. The complete per-compound classification for all 588 entries, listing formula, Materials Project identifier, space group, dominant-pair ICOBI/ICOHP, and the normalized ICOHP/ICOBI antibonding population, is reproduced by bonding-type category in Sections S5.1–S5.5.

S5.1 Metallic Compounds and Elements

Table S7: Bonding-type classification: composition-based heuristic vs. ICOHP/ICOBI-based classification ($n = 588$).

Heuristic \ ICOHP/ICOBI	metallic	ionic	covalent	mixed	unclassified
metallic	193	0	0	0	0
ionic	0	48	2	3	0
covalent	0	14	35	16	0
unclassified	179	36	19	43	0
Total	372	98	56	62	0

Concordance on the subset where the composition-based heuristic renders a verdict: $276/311 = 88.7\%$. The remaining 277 compounds the heuristic leaves “unclassified” (mostly alkali/alkaline-earth $\times \{\text{N,P,S}\}$ pairs outside a fixed anion list) are resolved by the ICOHP/ICOBI-based classification into 179 metallic, 36 ionic, 19 covalent, and 43 mixed (Zintl-type).

Table S8: Metallic compounds and elements (`is_metal=True`, observed ICOBI 0.0000 to 2.9506) — formula, mp-id, space group, dominant (first-shell) bond ICOBI, dominant bond ICOHP, normalized ICOHP and ICOBI antibonding populations ($\Delta E = 1.0$ eV).

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ag*	mp-124	Fm-3m	0.1545	-1.2496	0.0000	0.0000
Ag ₃ As	mp-1183193	P6 ₃ /mmc	0.2236	-1.3102	0.0000	0.0000
Ag ₃ Pt*	mp-12065	Pm-3m	0.2113	-1.5965	0.0033	0.0026
Ag ₃ Sb*	mp-2273	Pmm2	0.2406	-1.5221	0.0005	0.0000
AgI*	mp-22915	Pm-3m	0.2126	-1.1434	0.0699	0.0000
AgN*	mp-1091417	Cmce	0.4624	-3.4181	0.0088	0.0030
AgN	mp-1009766	F-43m	0.4854	-3.5762	0.0277	0.0173
AgN	mp-1424734	Ama2	0.8238	-5.9446	0.0107	0.0025
AgN	mp-1428312	Pm-3m	0.2177	-1.4991	0.0083	0.0041
AgPb ₃	mp-1183256	I4/mmm	0.2181	-1.2748	0.0000	0.0000
AgPd	mp-1183222	P-6m2	0.1929	-1.4704	0.0292	0.0199
Al*	mp-134	Fm-3m	0.2270	-1.4193	0.0000	0.0000
Al ₁₂ Mo*	mp-550	Im-3	0.3238	-1.5619	0.0000	0.0000
Al ₁₂ W*	mp-11227	Im-3	0.5027	-2.7492	0.0000	0.0000
Al ₂ W*	mp-12524	P6 ₃ /222	0.4364	-2.4880	0.0000	0.0000
Al ₃ Ag	mp-1183130	Pm-3m	0.2444	-1.4593	0.0000	0.0000
Al ₃ As	mp-1183129	Pm-3m	0.2253	-1.3902	0.0001	0.0000
Al ₃ Os ₂ *	mp-16521	I4/mmm	0.3022	-1.6476	0.0037	0.0000
Al ₄ Si	mp-1228120	R-3m	0.2614	-2.0375	0.0087	0.0030
AlAg	mp-1228901	P-6m2	0.2436	-1.7117	0.0000	0.0000
AlAg ₂	mp-985805	Cmcm	0.2723	-1.5388	0.0000	0.0000
AlNi*	mp-1487	Pm-3m	0.2676	-1.9848	0.0016	0.0023
AlSb	mp-1023918	Cmcm	0.2647	-1.2802	0.0000	0.0000
As*	mp-158	Cmce	0.8848	-3.9683	0.0103	0.0000
As ₃ W ₂ *	mp-11696	C2/m	0.7072	-4.3348	0.0000	0.0000
AsPd ₂ *	mp-1080831	P-62m	0.3410	-2.4185	0.0077	0.0157
AsRh*	mp-22079	Pnma	0.3323	-1.8639	0.0449	0.0158
AsRh ₂ *	mp-1102364	Pnma	0.2438	-1.3778	0.1158	0.0189
Au*	mp-81	Fm-3m	0.1962	-1.3381	0.0000	0.0000
B ₃ Ir ₂	mp-1228647	C2/m	0.4473	-3.3600	0.0047	0.0011
BPd ₅	mp-1228665	I4/mmm	0.4754	-3.9868	0.0971	0.0181
BPd ₅	mp-1228665	I4/mmm	0.4754	-3.9924	0.0973	0.0181
BW*	mp-7832	I4 ₁ /amd	0.4277	-3.9556	0.0000	0.0000

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
Ba*	mp-122	Im-3m	0.0002	-0.0122	8.2400	2.9573
Ba ₃ Ca	mp-1183342	I4/mmm	0.0000	0.0000	—	—
Ba ₃ N ₂	mp-1013528	Pm-3m	0.0763	-0.6612	0.1655	0.0007
BaAu ₂ *	mp-30363	P6/mmm	0.6152	-3.0712	0.0144	0.0034
BaF ₃	mp-1183303	I4/mmm	0.0350	-0.5277	0.2805	0.2537
BaMg ₃	mp-1016218	Pmm2	0.0023	-0.0755	5.4796	1.1861
BaN ₂ *	mp-1102371	Cc	0.1605	-0.4471	0.1039	0.0214
BaSn ₅ *	mp-571353	P6/mmm	0.2221	-0.8507	0.0456	0.0042
Be*	mp-87	P6 ₃ /mmc	0.2033	-1.8975	0.0000	0.0000
Be ₁₂ Mo*	mp-1475	I4/mmm	0.2043	-1.9103	0.0000	0.0000
Be ₁₂ Pd*	mp-12666	I4/mmm	0.2078	-1.9466	0.0000	0.0000
Be ₂ Cu*	mp-2031	Fd-3m	0.3616	-3.0863	0.0000	0.0001
Be ₂ Nb ₃ *	mp-11272	P4/mbm	0.4200	-3.5436	0.0000	0.0000
Be ₂ V*	mp-11281	P6 ₃ /mmc	0.3658	-2.7859	0.0000	0.0000
BeAu ₃	mp-983539	Pm-3m	0.2230	-1.9129	0.0000	0.0030
BeCu*	mp-2323	Pm-3m	0.1984	-1.8359	0.0000	0.0000
BeF ₂	mp-975644	P6/mmm	0.1548	-1.6978	0.0863	0.0000
Bi ₂ O ₃ *	mp-558953	P2 ₁ /c	0.9517	-6.7444	0.0328	0.0038
Bi ₂ O ₃	mp-1383505	P1	0.9506	-7.1985	0.0609	0.0000
C*	mp-48	P6 ₃ /mmc	1.2195	-11.5387	0.0000	0.0000
Ca*	mp-21	Im-3m	0.0025	-0.0304	3.2206	3.3021
Ca ₃ N ₂	mp-1013524	Pm-3m	0.1107	-0.3623	0.0090	0.0000
Ca ₃ Zn	mp-1039229	I4/mmm	0.1222	-0.3874	0.2004	0.0086
CaAl ₄ *	mp-1749	I4/mmm	0.4595	-1.8701	0.0086	0.0002
CaGa ₂ *	mp-992	P6/mmm	0.9791	-3.8887	0.0219	0.0000
CaGa ₄ *	mp-2461	C2/m	0.4641	-2.1536	0.0084	0.0001
CaGe ₂	mp-643542	C2/m	0.7379	-3.1645	0.0337	0.0038
CaN*	mp-1058549	Fm-3m	0.1201	-0.6934	0.0450	0.0656
CaNi*	mp-1059707	Pm-3m	0.0678	-0.1910	0.5038	0.0500
CaO	mp-1181903	Fd-3m	0.0725	-0.5516	0.1865	0.0756
CaSn*	mp-2450	Cmcm	0.7044	-2.1965	0.0629	0.0276
Cd*	mp-94	P6 ₃ /mmc	0.2423	-1.6119	0.0000	0.0000
CdIn ₃	mp-1183674	Fm-3m	0.2417	-1.4735	0.0000	0.0000
CdPd*	mp-1696	P4/mmm	0.2022	-1.5904	0.0000	0.0004
CdSb ₃	mp-1183739	P6 ₃ /mmc	0.2091	-1.1803	0.0378	0.0058
Co*	mp-102	Fm-3m	0.2680	-1.9150	0.0189	0.0373
CoB*	mp-20857	Pnma	0.4568	-4.5928	0.0099	0.0000
CoGe*	mp-21237	C2/m	0.4261	-2.9354	0.0002	0.0022
CoRe	mp-1225995	P-6m2	0.3611	-2.2350	0.0086	0.0082
CoSe ₂	mp-1390196	Fd-3m	0.7048	-4.3267	0.0348	0.0079
CoSn*	mp-20536	P6/mmm	0.4900	-2.7764	0.0035	0.0023
Cr*	mp-90	Im-3m	0.2487	-0.6245	0.0115	0.0049
Cr ₂ C	mp-1226378	P-3m1	0.5270	-3.0803	0.0084	0.0000
Cr ₅ Si ₃ *	mp-7506	I4/mcm	0.4031	-3.1831	0.0057	0.0003
CrMo	mp-1226200	Cmmm	0.3600	-1.3679	0.0081	0.0026
CrRe ₃	mp-1183703	Pm-3m	0.3358	-1.4730	0.0060	0.0000
CrW	mp-1226162	Cmmm	0.5827	-3.4455	0.0000	0.0005
Cs*	mp-1	Im-3m	0.0556	-0.1230	0.1696	0.0693
CsO ₂ *	mp-1441	I4/mmm	1.0942	-11.4627	0.1143	0.1436
CsO ₂	mp-1096936	Cmmm	1.1298	-11.0556	0.1197	0.0872
CsRb	mp-1225895	Cmmm	0.0628	-0.1628	0.1759	0.0961
Cu*	mp-30	Fm-3m	0.0456	-0.2488	0.1641	0.0946
Cu ₂ S*	mp-12087	Fm-3m	0.1495	-0.9496	0.1576	0.0012
Cu ₂ S*	mp-553942	P6 ₃ /mmc	0.0830	-0.4779	0.1504	0.0087
Cu ₃ Ge*	mp-19724	Pmmn	0.1369	-0.8340	0.0963	0.0410
Cu ₃ Te ₂	mp-1225676	P4mm	0.1480	-0.8199	0.1597	0.0663
Cu ₇ Te ₄ *	mp-624307	P3m1	0.1323	-0.6945	0.1434	0.0510

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
CuH	mp-1225705	P-3m1	0.0446	-0.1622	0.1097	0.0037
CuNi	mp-1184069	P-6m2	0.1198	-0.4298	0.4478	0.0732
Fe*	mp-13	Im-3m	0.1417	-0.5370	0.0922	0.0794
FeS*	mp-505531	P4/nmm	0.4967	-2.5883	0.0736	0.0024
FeTe ₂ *	mp-1102002	Pa-3	0.4279	-1.7861	0.0401	0.0000
Ga*	mp-142	Cmce	0.4308	-2.6378	0.0000	0.0000
Ga ₂ Au ₇ *	mp-571598	P-62m	0.2592	-1.8968	0.0000	0.0000
Ga ₃ Hg	mp-1184289	P6 ₃ /mmc	0.2216	-1.4162	0.0000	0.0000
Ga ₃ Pb	mp-1184425	I4/mmm	0.1929	-1.0646	0.0042	0.0000
Ga ₃ Re*	mp-1188393	P4 ₂ /mnm	0.4857	-2.3451	0.0000	0.0000
GaSb	mp-1224785	I-4m2	0.2847	-1.4795	0.0030	0.0000
Ge*	mp-32	Fd-3m	0.8956	-4.6450	0.0015	0.0000
Ge ₂ Os*	mp-10032	C2/m	0.4042	-2.1066	0.0069	0.0003
Ge ₃ Mo ₅ *	mp-17094	I4/mcm	0.4229	-3.3674	0.0068	0.0009
Ge ₃ O	mp-1184665	I4/mmm	0.1453	-0.8722	0.0091	0.0027
Ge ₄ Rh*	mp-1104286	P3 ₁ 21	0.3355	-1.8036	0.0034	0.0006
GeAs*	mp-7591	I4mm	0.4634	-2.4402	0.0063	0.0014
GeSb	mp-9935	Fm-3m	0.4373	-2.1513	0.0101	0.0039
H ₃ Pd ₄	mp-1224413	R-3m	0.1466	-1.0855	0.1005	0.0202
H ₄ N	mp-1238814	P-43m	0.6754	-6.0262	0.0498	0.0006
HC ₂	mp-995184	C2/m	1.2737	-11.4891	0.0000	0.0025
HC ₃	mp-995200	P2 ₁ /m	1.2493	-11.5739	0.0000	0.0000
HC ₆	mp-995202	P-1	1.2320	-11.5624	0.0000	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.3121	-1.1612	0.0000	0.0000
Hf ₃ Pb	mp-1184772	I4/mmm	0.3286	-1.3613	0.0000	0.0000
HfCd*	mp-1007758	P4/nmm	0.3741	-1.6812	0.0000	0.0000
HfCr ₂ *	mp-569624	P6 ₃ /mmc	0.4565	-1.1713	0.0000	0.0000
HfCu ₃	mp-1184704	I4/mmm	0.1762	-0.7182	0.0196	0.0000
HfMg ₁₅	mp-1023478	P-6m2	0.0440	-0.1164	1.0039	0.4374
HfMo	mp-1224314	Cmmm	0.3368	-1.2722	0.0000	0.0000
HfTc ₂ *	mp-1095669	P6 ₃ /mmc	0.3775	-1.4429	0.0098	0.0012
HfW ₂	mp-1224230	Fmmm	0.4943	-2.5963	0.0000	0.0000
Hg*	mp-10861	P6/mmm	0.1622	-0.9652	0.0000	0.0000
Hg ₃ Pb	mp-1184700	P6 ₃ /mmc	0.2256	-1.4191	0.0000	0.0000
Hg ₃ Sb	mp-977242	P6 ₃ /mmc	0.2146	-1.3610	0.0001	0.0003
Hg ₄ Pt*	mp-936	Im-3m	0.5477	-3.5588	0.0000	0.0005
Hg ₅ Pd ₂ *	mp-30721	P4/mbm	0.2364	-1.7271	0.0000	0.0002
IN ₂	mp-1097011	I4/mcm	2.9506	-23.1289	0.0177	0.0275
In*	mp-1055994	I4/mmm	0.2177	-1.1562	0.0000	0.0000
In ₂ Te ₃ *	mp-1105025	I4mm	0.6032	-2.6916	0.0224	0.0090
In ₃ Ir*	mp-630976	P4 ₂ /mnm	0.5910	-3.4701	0.0000	0.0000
In ₇ Cu ₁₀ *	mp-646039	C2/m	0.1416	-0.8072	0.0230	0.0039
InNi*	mp-20997	Pm-3m	0.2785	-1.7672	0.0081	0.0156
InSe ₂	mp-1232036	P4/nmm	0.2771	-1.4780	0.0349	0.0038
InSn ₄	mp-1223653	Cmmm	0.3867	-2.1185	0.0000	0.0000
Ir*	mp-101	Fm-3m	0.3469	-2.4130	0.0317	0.0274
Ir ₃ Ru	mp-974358	I4/mmm	0.3767	-2.6659	0.0317	0.0198
Ir ₄ W	mp-1223666	R-3m	0.3980	-2.8035	0.0422	0.0434
IrPd	mp-1223675	R-3m	0.3508	-2.9933	0.0756	0.0384
K*	mp-10157	Fm-3m	0.0430	-0.0808	0.0923	0.0499
K ₂ O	mp-1223784	P4/mmm	0.1375	-0.3763	0.0831	0.0000
K ₃ Rb	mp-976115	P6 ₃ /mmc	0.0431	-0.0882	0.1130	0.0889
K ₅ As ₄	mp-684799	P-1	0.9859	-3.9306	0.0452	0.0189
KMg ₁₅	mp-1023498	P-6m2	0.0006	-0.0094	4.8845	1.9656
KN ₂ *	mp-1071868	Cmc2 ₁	1.0490	-7.9889	0.0202	0.0000
KN ₂	mp-1180757	Cmcm	1.0210	-7.4152	0.0411	0.0000
KN ₃	mp-636056	I4/mcm	2.8711	-22.8747	0.0090	0.0107

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
KP [*]	mp-1058315	R-3m	0.1254	-0.4533	0.0190	0.0442
KP	mp-1057787	Immm	0.1070	-0.4621	0.0170	0.0218
KSn ₃	mp-1185151	P6 ₃ /mmc	0.4006	-1.5431	0.0179	0.0027
Li [*]	mp-1018134	R-3m	0.0886	-0.6931	0.0000	0.0000
Li ₁₃ Sn ₅ [*]	mp-30769	P-3m1	0.6069	-2.7881	0.0000	0.0000
Li ₃ Ag	mp-976408	I4/mmm	0.1084	-1.0444	0.0000	0.0000
Li ₃ Ca	mp-976075	I4/mmm	0.1652	-0.6223	0.0000	0.0000
Li ₃ Hg [*]	mp-1646	Fm-3m	0.1488	-1.2194	0.0000	0.0000
LiB [*]	mp-1001835	P6 ₃ /mmc	1.1571	-9.9815	0.0000	0.0000
LiRh ₃ [*]	mp-1001787	Imm2	0.1776	-1.5461	0.0814	0.0192
LiTi ₃	mp-1185253	I4/mmm	0.2370	-1.2989	0.0000	0.0000
Mg [*]	mp-153	P6 ₃ /mmc	0.0000	0.0000	—	—
Mg ₁₅ B	mp-1023515	P-6m2	0.0280	-0.1152	0.9423	0.5909
Mg ₁₅ Ni	mp-1023622	P-6m2	0.0020	-0.0213	3.2054	2.1119
Mg ₂ Ge [*]	mp-250	Fd-3m	0.1806	-0.5039	0.1144	0.0306
Mg ₂ Ni [*]	mp-2137	P6 ₃ /222	0.0818	-0.2478	0.6457	0.3329
Mg ₃ Al	mp-978271	Pm-3m	0.0812	-0.3224	0.1540	0.0681
Mg ₃ Ru ₂ [*]	mp-625	P4 ₃ /2	0.5409	-1.6132	0.1816	0.0936
Mg ₃ Zr	mp-1094466	P6 ₃ /mmc	0.2889	-0.3975	0.0887	0.0141
Mg ₅ Sc	mp-1185837	R32	0.1000	-0.1246	0.2961	0.0068
Mg ₅ Te	mp-1185828	R32	0.0108	-0.0917	1.0311	0.5406
MgAl ₂ [*]	mp-1102064	I4 ₁ /amd	0.3363	-1.6067	0.0037	0.0000
MgCd [*]	mp-11307	Pmma	0.4362	-2.0470	0.0026	0.0006
MgIn ₂ [*]	mp-1077016	Fd-3m	0.4220	-1.8995	0.0118	0.0002
MgPb ₂	mp-2016042	C2/m	0.2720	-1.0252	0.0628	0.0086
MgPd ₃ [*]	mp-7753	Pm-3m	0.2542	-1.7837	0.0264	0.0304
MgSn	mp-1094669	Pm-3m	0.2238	-0.7050	0.1000	0.0334
MgSn	mp-1094801	P4/mmm	0.3134	-1.2760	0.1099	0.0281
MgTi	mp-1094346	P2 ₁ /m	0.4231	-0.7659	0.0009	0.0000
MgTl ₃	mp-978262	P6 ₃ /mmc	0.2760	-1.3397	0.0002	0.0000
Mn [*]	mp-35	I-43m	—	-0.6082	0.0411	0.0047
Mn ₄ B	mp-1221769	F222	0.3919	-2.3186	0.0085	0.0007
Mn ₄ Ga	mp-1221812	R-3m	0.3290	-2.4448	0.0073	0.0017
MnAl ₁₂ [*]	mp-1104165	Im-3	0.2798	-1.6283	0.0000	0.0000
MnO ₂ [*]	mp-510408	P4 ₂ /mnm	0.4433	-3.0527	0.0548	0.0603
Mo [*]	mp-129	Im-3m	0.3113	-1.1059	0.0139	0.0005
Mo ₂ C	mp-1221498	P-3m1	0.5408	-3.5050	0.0057	0.0002
Mo ₃ Se ₄ [*]	mp-21021	R-3	0.4924	-2.4124	0.0151	0.0004
MoPd ₂	mp-1206704	Immm	0.5227	-1.7521	0.0013	0.0000
MoRu	mp-1221386	P-6m2	0.3772	-1.3952	0.0182	0.0004
N ₂	mp-1059834	Imma	1.3533	-15.9343	0.0159	0.0010
NO	mp-1180334	P2 ₁₂ /12 ₁	1.8010	-21.3123	0.0427	0.0039
Na [*]	mp-10172	P6 ₃ /mmc	0.0470	-0.0839	0.1168	0.0901
Na ₂ Cl	mp-1077015	Cmmm	0.0377	-0.1951	0.0901	0.0029
Na ₂ Cl	mp-990084	R-3m	0.0479	-0.3527	0.0618	0.0000
Na ₃ Cl	mp-1064484	P4/mmm	0.0297	-0.2254	0.0859	0.0249
Na ₃ Hg	mp-1186050	I4/mmm	0.1243	-0.3788	0.0010	0.0000
Na ₃ In	mp-864758	I4/mmm	0.1784	-0.3865	0.0055	0.0000
Na ₃ Mg	mp-982637	I4/mmm	0.0436	-0.0860	0.1585	0.0598
Na ₃ Si ₃₄	mp-1221235	R-3m	0.8887	-4.3194	0.0033	0.0052
Nb [*]	mp-75	Im-3m	0.3268	-1.1439	0.0000	0.0000
Nb ₃ Ir [*]	mp-1458	Pm-3n	0.4144	-2.0103	0.0000	0.0002
Nb ₃ Ni	mp-999394	I4/mmm	0.4256	-1.5440	0.0000	0.0002
Nb ₄ Rh	mp-1220441	Fmmm	0.3405	-1.1843	0.0002	0.0000
NbCd ₃ [*]	mp-11309	Pm-3m	0.2671	-1.6744	0.0000	0.0000
NbGa ₃ [*]	mp-1973	I4/mmm	0.3104	-1.7262	0.0000	0.0000
NbO [*]	mp-2692	Fm-3m	0.2933	-2.2812	0.0292	0.0023

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
NbO	mp-634965	P4/mmm	0.9184	-5.5839	0.0000	0.0000
NbP [*]	mp-9830	I4 ₁ /amd	0.3299	-1.7803	0.0001	0.0000
NbP	mp-1220317	Cmmm	0.3238	-1.7805	0.0084	0.0004
NbRh ₂	mp-1220414	Pm	0.3585	-1.4459	0.0440	0.0023
NbS ₂	mp-774873	P6 ₃ /mmc	0.6390	-3.1946	0.0049	0.0000
NbZn ₁₆ [*]	mp-640041	Cmcm	0.2534	-1.6346	0.0000	0.0000
Ni [*]	mp-23	Fm-3m	0.0947	-0.4204	0.3799	0.2037
Ni ₃ Mo [*]	mp-11506	Pmmn	0.2915	-1.0002	0.1370	0.0141
NiB [*]	mp-14019	Cmcm	0.6996	-6.1857	0.0198	0.0093
NiHg ₄ [*]	mp-570835	Im-3m	0.3125	-2.2179	0.0000	0.0004
NiS [*]	mp-1547	R3m	0.2630	-1.5895	0.1140	0.0342
NiSb ₃ [*]	mp-1106023	Im-3	0.3169	-1.4693	0.0547	0.0298
Os [*]	mp-49	P6 ₃ /mmc	0.2567	-1.1579	0.0538	0.0135
OsC [*]	mp-7142	P-6m2	0.4688	-3.0941	0.0402	0.0068
OsC	mp-1009537	Pm-3m	0.3353	-2.0943	0.0446	0.0000
OsC	mp-1009539	Fm-3m	0.4618	-3.1760	0.0828	0.0058
OsC	mp-999308	P6 ₃ /mmc	0.4521	-3.1583	0.0525	0.0226
PW ₃	mp-1209828	I-42m	0.4114	-2.7903	0.0000	0.0000
Pb [*]	mp-20483	Fm-3m	0.1943	-0.8519	0.0276	0.0000
Pb ₂ Au [*]	mp-22795	I4/mcm	0.4524	-3.2812	0.0000	0.0000
PbAu ₂ [*]	mp-19871	Fd-3m	0.3048	-2.3586	0.0000	0.0000
PbAu ₃	mp-982246	Fm-3m	0.1941	-1.4564	0.0112	0.0022
Pd [*]	mp-2	Fm-3m	0.1631	-1.2162	0.1264	0.0877
Pt [*]	mp-126	Fm-3m	0.2367	-1.8089	0.0927	0.0819
Rb [*]	mp-70	Im-3m	0.0590	-0.1180	0.1307	0.0377
Rb ₂ Hg ₇ [*]	mp-31474	P-3m1	0.1920	-0.9499	0.0000	0.0000
Rb ₃ In	mp-974948	Pm-3m	0.1129	-0.2266	0.0341	0.0000
Rb ₃ Li	mp-974938	Pm-3m	0.0991	-0.3104	0.0185	0.0000
Rb ₃ P ₁₁	mp-1173444	C2/m	0.8390	-4.2144	0.0355	0.0027
RbP [*]	mp-1179688	C2/m	0.1031	-0.4288	0.0138	0.0467
RbP	mp-1058499	Immm	0.1277	-0.4304	0.0099	0.0324
Re [*]	mp-8	P6 ₃ /mmc	0.3162	-1.3803	0.0205	0.0030
Re ₂ Si [*]	mp-669547	P2 ₁ /c	0.3849	-1.8091	0.0190	0.0006
Re ₃ Ge ₇ [*]	mp-29141	Cmcm	0.4553	-2.2942	0.0022	0.0002
Re ₃ N ₅	mp-1080194	Imm2	0.8324	-5.2882	0.0269	0.0001
Re ₃ Pd	mp-862586	P6 ₃ /mmc	0.3788	-1.7449	0.0020	0.0011
Re ₅ Si ₃ [*]	mp-1105959	I4/mcm	0.3379	-2.7922	0.0251	0.0018
ReC [*]	mp-1079494	P6 ₃ /mmc	0.4741	-3.3056	0.0211	0.0022
ReC	mp-1009731	Pm-3m	0.3588	-2.2826	0.0149	0.0003
Rh [*]	mp-74	Fm-3m	0.1620	-0.7081	0.1852	0.0924
Ru [*]	mp-33	P6 ₃ /mmc	0.2375	-0.9478	0.0875	0.0211
RuC [*]	mp-10030	P-6m2	0.4770	-2.9147	0.0418	0.0078
RuC	mp-1009787	Fm-3m	0.4752	-2.9734	0.0862	0.0067
RuRh ₃	mp-974335	Pm-3m	0.2074	-0.8611	0.1382	0.0326
SN	mp-1078816	P2 ₁ /c	0.2823	-1.7903	0.0359	0.0006
SN	mp-1173453	P2 ₁	1.5195	-10.9244	0.0458	0.0000
SN	mp-684642	P2 ₁	1.3504	-13.4315	0.0333	0.0038
Sb [*]	mp-104	R-3m	0.4398	-1.7202	0.0425	0.0063
Sb ₂ S ₅	mp-1359089	P2 ₁ /c	1.0848	-6.8891	0.0690	0.0097
Sc [*]	mp-67	P6 ₃ /mmc	0.2110	-0.6596	0.0000	0.0000
Sc ₅ Ge ₃ [*]	mp-17190	P6 ₃ /mcm	0.3162	-1.5145	0.0000	0.0000
ScAg ₂ [*]	mp-1018128	I4/mmm	0.2014	-1.5516	0.0000	0.0000
ScOs ₃	mp-981384	Pm-3m	0.2618	-1.1903	0.0252	0.0106
ScTc ₃	mp-867262	Fm-3m	0.3043	-1.0739	0.0080	0.0010
ScTe [*]	mp-10026	P6 ₃ /mmc	0.3365	-1.5673	0.0000	0.0000
Si ₃ Pt	mp-972763	P6 ₃ /mmc	0.4284	-2.6580	0.0034	0.0000
Si ₃ W ₅ [*]	mp-31219	I4/mcm	0.4312	-2.5598	0.0000	0.0000

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
SiAu ₃	mp-1186998	Fm-3m	0.1974	-1.4121	0.0156	0.0000
SiO ₂	mp-556044	P-31c	0.7437	-6.4399	0.0117	0.0000
SiO ₂	mp-638049	F4 ₋₁₃₂	0.6594	-6.0556	0.0488	0.0000
SiRh*	mp-2287	P2 _{-1/c}	0.2801	-1.6004	0.0243	0.0009
SiTc ₃	mp-972784	P6 _{-3/mmc}	0.3385	-1.4573	0.0257	0.0020
SiTe ₂ *	mp-2755	P-3m1	0.5658	-2.7523	0.0695	0.0130
Sn*	mp-623511	I4/mmm	0.1989	-0.8803	0.0235	0.0006
Sn*	mp-117	Fd-3m	0.8707	-3.6930	0.0001	0.0004
Sn ₃ S	mp-1187038	Fm-3m	0.2241	-1.1171	0.0176	0.0000
Sn ₃ Se	mp-1187055	Fm-3m	0.2108	-1.0119	0.0214	0.0027
Sn ₇ Ir ₅ *	mp-20466	I4mm	0.5375	-3.2800	0.0000	0.0000
Sn ₇ Pd ₅	mp-1219006	C2	0.3765	-2.7926	0.0000	0.0004
SnP*	mp-475	Fm-3m	0.4347	-2.3226	0.0105	0.0058
SnPb ₃	mp-1187118	P6 _{-3/mmc}	0.1988	-0.8714	0.0218	0.0000
SnPt*	mp-19856	P6 _{-3/mmc}	0.4612	-3.2535	0.0000	0.0000
Sr*	mp-139	P6 _{-3/mmc}	0.0009	-0.0171	4.9050	5.3659
Sr ₂ As*	mp-15697	I4/mmm	0.0787	-0.2746	0.2097	0.0267
Sr ₃ N ₂	mp-1013529	Pm-3m	0.1064	-0.4148	0.0858	0.0000
Sr ₃ Sn	mp-1187136	Pm-3m	0.0501	-0.2374	0.4377	0.0368
Sr ₅ Sn ₃ *	mp-17720	I4/mcm	0.8182	-2.1022	0.1796	0.0000
SrAl ₂ *	mp-1638	Fd-3m	0.4263	-1.6649	0.0004	0.0000
SrB ₂	mp-1008755	P6/mmm	0.9937	-5.2175	0.0052	0.0007
SrN*	mp-1058586	Fm-3m	0.1230	-0.6070	0.0462	0.0849
SrO ₁₀	mp-1208655	P4/mmm	0.8181	-6.3571	0.1157	0.0000
SrSi*	mp-2661	Cmcm	0.8077	-2.8883	0.0564	0.0097
SrTi ₃	mp-1206636	Pm-3m	0.2857	-1.2068	0.0201	0.0007
Ta*	mp-50	Im-3m	0.3226	-1.2794	0.0000	0.0000
Ta ₄ C ₃	mp-1218000	R3m	0.5265	-3.5440	0.0000	0.0000
Ta ₅ Ge ₃ *	mp-17593	I4/mcm	0.3556	-2.6739	0.0000	0.0000
Ta ₇ Co ₆ *	mp-1104548	R-3m	0.4891	-3.0211	0.0018	0.0000
TaBe ₁₂ *	mp-567842	I4/mmm	0.2080	-1.8875	0.0000	0.0000
TaN*	mp-1497	P6/mmm	0.4241	-2.9806	0.0009	0.0000
TaN	mp-1009831	Pm-3m	0.3185	-2.1377	0.0002	0.0002
TaN	mp-1009833	F-43m	0.6451	-4.4476	0.0000	0.0006
TaP*	mp-1067587	I4 _{-1md}	0.4762	-2.7480	0.0000	0.0000
TaP	mp-1187244	P-6m2	0.4792	-2.7509	0.0000	0.0000
TaRe	mp-1217894	Cmmm	0.3311	-1.5114	0.0026	0.0003
TaSb ₂ *	mp-11697	C2/m	0.4899	-2.0334	0.0002	0.0000
Tc*	mp-113	P6 _{-3/mmc}	0.3130	-1.2061	0.0240	0.0006
Tc ₃ Mo	mp-1187395	Pm-3m	0.3332	-1.2727	0.0068	0.0014
Tc ₃ Os	mp-862779	P6 _{-3/mmc}	0.3182	-1.2186	0.0252	0.0052
Tc ₃ W	mp-1187398	P6 _{-3/mmc}	0.4723	-2.3705	0.0000	0.0000
TcW	mp-1217337	Cmmm	0.6035	-3.6836	0.0000	0.0002
Te ₂ Au*	mp-567525	C2/m	0.5036	-3.0627	0.0497	0.0327
Te ₂ Ir*	mp-1551	C2/m	0.7621	-4.4689	0.0230	0.0046
Te ₂ Pd*	mp-782	P-3m1	0.5813	-3.5060	0.0291	0.0200
Te ₂ Pd ₇	mp-1217443	R-3m	0.2769	-1.8731	0.0363	0.0145
Ti*	mp-46	P6 _{-3/mmc}	0.3010	-0.7536	0.0000	0.0000
Ti ₂ Cu*	mp-742	I4/mmm	0.3374	-0.7591	0.0016	0.0000
Ti ₃ Ge*	mp-1189044	I-4	0.3243	-1.6051	0.0000	0.0000
Ti ₃ In*	mp-866046	P6 _{-3/mmc}	0.3128	-1.1764	0.0000	0.0000
Ti ₃ Nb	mp-1217091	Amm2	0.3774	-1.2750	0.0000	0.0000
Ti ₃ Pt*	mp-2339	Pm-3n	0.3632	-1.7698	0.0000	0.0000
TiBe ₁₂ *	mp-1104067	I4/mmm	0.2186	-1.9904	0.0000	0.0000
TiMo ₂	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiMo ₂	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiNi*	mp-1048	P2 _{-1/m}	0.2818	-0.8685	0.0059	0.0000

Formula	mp-id	Space group	ICOB	ICOHP (eV)	ICOHP antibonding	ICOB antibonding
TiNi	mp-1067248	Cmcm	0.2798	-0.8586	0.0060	0.0000
TiO*	mp-1071163	P-62m	0.3594	-2.4547	0.0015	0.0000
TiSi ₂ *	mp-1077503	Cmcm	0.4599	-3.0700	0.0000	0.0000
TiSi ₂	mp-570991	Immm	0.2659	-1.2939	0.0000	0.0000
TiW	mp-1216621	Cmmm	0.6568	-3.6207	0.0000	0.0000
Tl*	mp-82	P6 ₃ /mmc	0.1905	-1.0463	0.0000	0.0000
Tl ₃ Au	mp-1080569	Pm-3n	0.2291	-1.2850	0.0000	0.0000
Tl ₃ Ga	mp-1187539	Fm-3m	0.2090	-1.0914	0.0000	0.0000
Tl ₄ Sn	mp-1216560	R-3m	0.2039	-1.3837	0.0010	0.0008
TlAs	mp-1007770	F-43m	0.7362	-3.5581	0.0000	0.0002
TlAu ₃	mp-1187508	Pm-3m	0.2130	-1.5377	0.0000	0.0000
TiC*	mp-567465	Fm-3m	0.3776	-2.4265	0.0000	0.0010
TiC	mp-1058887	P4/mmm	0.4837	-3.1370	0.0000	0.0003
TiN*	mp-1017982	P6 ₃ mc	0.6066	-3.6824	0.0023	0.0000
TiN	mp-1002222	Fm-3m	0.3446	-2.1516	0.0029	0.0000
TiN	mp-1007778	F-43m	0.6105	-3.6815	0.0017	0.0000
V*	mp-146	Im-3m	0.3178	-0.8777	0.0000	0.0000
VC ₀₃	mp-1095057	P6 ₃ /mmc	0.3730	-2.5141	0.0012	0.0023
VN	mp-1017532	P6 ₃ /mmc	0.4293	-2.6954	0.0000	0.0000
VNi ₂ *	mp-11531	Immm	0.4769	-1.4029	0.0918	0.0008
W*	mp-91	Im-3m	0.4372	-2.5600	0.0000	0.0001
W ₉ C ₄	mp-684989	R32	0.6031	-4.5942	0.0000	0.0000
WBr ₅	mp-1436789	P-1	0.8263	-4.3695	0.0774	0.0061
WC*	mp-13136	Fm-3m	0.5508	-4.2240	0.0012	0.0009
WC	mp-1008630	Pm-3m	0.4127	-2.9246	0.0000	0.0010
WC	mp-1008635	F-43m	0.8710	-6.2790	0.0000	0.0000
WO ₂ *	mp-1524394	P2 ₁ /c	0.6713	-5.7408	0.0039	0.0000
Y*	mp-112	P6 ₃ /mmc	0.2374	-0.8682	0.0000	0.0000
Y ₃ Co*	mp-1105598	Pnma	0.4720	-2.6094	0.0000	0.0000
Y ₃ Co	mp-1189329	P2 ₁	0.4718	-2.6045	0.0000	0.0000
YAg ₃	mp-1187811	Fm-3m	0.1780	-1.2749	0.0000	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.2296	-1.8892	0.0000	0.0000
Zn	mp-2647117	Im-3m	0.1819	-1.3641	0.0000	0.0000
Zn ₁₃ Rh*	mp-13447	C2/m	0.1989	-1.5615	0.0000	0.0000
Zn ₂ Ag	mp-1093999	P6 ₃ /mmc	0.2898	-2.4762	0.0000	0.0000
Zn ₃ Au	mp-1188000	P6 ₃ /mmc	0.2275	-1.8469	0.0000	0.0000
Zn ₃ N ₂	mp-1524551	P6 ₃ /mmc	0.3888	-2.8616	0.0030	0.0000
Zn ₃ N ₂ *	mp-9460	Ia-3	0.4180	-3.1067	0.0036	0.0000
Zn ₇ Tc	mp-1206378	Fm-3m	0.2749	-1.7542	0.0000	0.0009
ZnAg*	mp-1912	Pm-3m	0.2035	-1.8265	0.0000	0.0000
ZnPb ₃	mp-1187949	Fm-3m	0.2392	-1.3227	0.0004	0.0004
ZnSn	–	P6 ₃ /mmc	0.4604	-3.0247	0.0000	0.0000
Zr*	mp-131	P6 ₃ /mmc	0.3223	-1.3170	0.0000	0.0000
Zr ₂ Au*	mp-2348	I4/mmm	0.3095	-2.0203	0.0000	0.0000
Zr ₃ Be	mp-1188016	I4/mmm	0.3655	-1.7390	0.0000	0.0000
Zr ₃ Fe*	mp-31205	Cmcm	0.4128	-1.4259	0.0000	0.0000
Zr ₃ Ge	mp-1188039	P6 ₃ /mmc	0.3460	-1.7196	0.0000	0.0000
Zr ₅ Pb ₃ *	mp-681992	P6 ₃ /mcm	0.3301	-1.9739	0.0000	0.0000
ZrBr*	mp-1065846	C2/m	0.3528	-2.0185	0.0000	0.0001
ZrGa ₃ *	mp-30685	I4/mmm	0.2933	-1.8532	0.0000	0.0000
ZrHg ₃ *	mp-1608	Pm-3m	0.2490	-1.9086	0.0000	0.0000
ZrMo ₂ *	mp-2049	Fd-3m	0.4671	-1.6863	0.0002	0.0000
ZrRh*	mp-2808	Pm-3m	0.3062	-1.4732	0.0094	0.0024
ZrTi	mp-1215236	P4/mmm	0.3777	-1.6592	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §).

S5.2 Ionic Compounds

Table S9: Ionic compounds (non-metallic, $\text{ICOBI} < 0.5315$, observed range 0.0444 to 0.5305) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
AgH	mp-1096804	P6 ₃ mc	0.2039	-1.5451	0.0000	0.0000
AgO*	mp-1079720	P2 ₁ /c	0.2604	-2.0408	0.0297	0.0024
AlH ₃ *	mp-1106356	Fd-3m	0.4074	-2.4374	0.0000	0.0000
B*	mp-160	R-3m	0.5203	-4.7960	0.0000	0.0000
B ₆ Pb	mp-1096825	Pm-3m	0.2353	-1.9376	0.0000	0.0000
Ba ₄ Pd	mp-1214438	Fd-3m	0.4601	-3.3443	0.1949	0.0075
BaCl ₂ *	mp-568662	Fm-3m	0.0749	-0.5008	0.2998	0.0066
BaCl ₂ *	mp-23199	Pnma	0.0637	-0.4308	0.3027	0.0019
BaF ₃ *	mp-1080821	P2 ₁ /m	0.2279	-0.7883	0.3169	0.0437
BaH ₂ *	mp-23715	Pnma	0.0615	-0.2536	0.1495	0.0000
BaO*	mp-1008500	P6 ₃ /mmc	0.0788	-0.6380	0.1468	0.0016
BaO	mp-1950989	P6 ₃ /mmc	0.1070	-0.4367	0.2225	0.0000
Be ₃ N ₂ *	mp-6977	P6 ₃ /mmc	0.4361	-4.2580	0.0000	0.0000
Be ₃ N ₂ *	mp-18337	Ia-3	0.4759	-4.7015	0.0000	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.3952	-3.8753	0.0000	0.0000
BeCl ₂	mp-1172909	I-43m	0.3697	-0.9454	0.8095	0.1761
BeF ₂ *	mp-15951	P3 ₁ 121	0.5269	-6.8585	0.0035	0.0000
BeO*	mp-7599	P4 ₂ /mnm	0.5148	-5.7846	0.0000	0.0000
BeO	mp-1778	F-43m	0.5232	-5.8507	0.0000	0.0000
Ca ₃ N ₂ *	mp-1047	R-3c	0.1781	-0.6015	0.1048	0.0000
Ca ₃ N ₂ *	mp-844	Ia-3	0.1820	-0.6211	0.0837	0.0000
CaCl ₂ *	mp-23214	Pnnm	0.1054	-0.7540	0.2048	0.0008
CaCl ₂	mp-1120739	Fm-3m	0.0814	-0.5708	0.3236	0.0068
CaF ₂ *	mp-10464	Pnma	0.0565	-0.5566	0.2424	0.0000
CaF ₂	mp-554355	P4/mmm	0.0591	-0.5440	0.1286	0.0000
CaO*	mp-1079707	Cmce	0.1008	-0.5793	0.1805	0.0000
CaO*	mp-2605	Fm-3m	0.1143	-0.7028	0.0897	0.0000
CdI ₂ *	mp-567259	P-3m1	0.4859	-2.4996	0.0508	0.0000
Cs ₂ S*	mp-1077814	P6 ₃ /mmc	0.0804	-1.0011	0.0682	0.0000
Cs ₂ Se*	mp-1011696	Pnma	0.0811	-0.6189	0.0861	0.0000
CsCl*	mp-573697	Fm-3m	0.0595	-0.3453	0.0490	0.0000
CsF*	mp-8455	Pm-3m	0.1224	-1.5102	0.0523	0.0008
CsH*	mp-632319	Pm-3m	0.0935	-1.2305	0.0678	0.0002
GeS	mp-12910	Cmcm	0.4348	-2.2462	0.0516	0.0000
GeTe*	mp-938	R3m	0.4101	-1.7409	0.0714	0.0021
H ₂ O*	mp-2739191	C222 ₁	0.4372	-4.6426	0.0087	0.0000
HI*	mp-697025	C2/c	0.4915	-2.7099	0.0522	0.0000
HgF ₂ *	mp-8177	Fm-3m	0.2320	-1.8744	0.1660	0.0000
I*	mp-23153	Cmce	0.0444	-0.1504	0.4981	0.0900
IF ₇ *	mp-27988	Aea2	0.5305	-5.5662	0.0568	0.0000
InBr*	mp-22870	Cmcm	0.3010	-1.3292	0.0929	0.0024
InCl*	mp-571636	Cmc2 ₁	0.1698	-0.8368	0.0921	0.0006
K ₂ O*	mp-971	Fm-3m	0.1009	-0.8814	0.0059	0.0000
K ₂ S*	mp-559278	P6 ₃ /mmc	0.1040	-0.4891	0.0071	0.0006
K ₂ S*	mp-1022	Fm-3m	0.1212	-0.4744	0.0000	0.0000
KCl*	mp-23289	Pm-3m	0.0510	-0.4078	0.1374	0.0000
KF ₃ *	mp-1544557	P2 ₁ 12 ₁ 12 ₁	0.3162	-0.9256	0.4749	0.3545
KF ₃	mp-2749823	P2 ₁	0.3119	-0.9244	0.4733	0.3514
KI*	mp-22898	Fm-3m	0.0893	-0.4549	0.0615	0.0000
Li ₂ O*	mp-1960	Fm-3m	0.2393	-2.0320	0.0000	0.0000
Li ₂ O	mp-755894	Pnma	0.2225	-2.0369	0.0000	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Li ₂ S*	mp-1125	Pnma	0.2847	-2.0194	0.0000	0.0000
Li ₂ S	mp-557142	P6 ₃ /mmc	0.3327	-2.1900	0.0000	0.0000
Li ₂ Se*	mp-2286	Fm-3m	0.3399	-2.1702	0.0000	0.0000
Li ₃ N*	mp-2341	P6 ₃ /mmc	0.2587	-1.8972	0.0000	0.0000
LiBr*	mp-23259	Fm-3m	0.2790	-2.0381	0.0000	0.0000
LiCl*	mp-22905	Fm-3m	0.2558	-2.0554	0.0000	0.0000
LiCl	mp-1185319	P6 ₃ mc	0.3879	-2.9688	0.0000	0.0000
LiF*	mp-1138	Fm-3m	0.1692	-1.9334	0.0001	0.0000
LiF	mp-1009009	Pm-3m	0.1237	-1.3411	0.0249	0.0000
MgCl ₂ *	mp-570259	P-3m1	0.1312	-0.9334	0.2905	0.0028
MgCl ₂ *	mp-23210	R-3m	0.1313	-0.9344	0.2883	0.0009
MgCl ₂	mp-569626	Ama2	0.1475	-1.0096	0.1898	0.0001
MgF ₂ *	mp-1072956	Pnnm	0.0901	-1.0192	0.1330	0.0000
MgF ₂	mp-1062842	Amm2	0.0808	-0.8465	0.1522	0.0000
MgF ₂	mp-560236	C2/m	0.1007	-1.0813	0.0878	0.0000
MgH ₂ *	mp-23711	Pbcn	0.1156	-0.5595	0.0617	0.0000
MgO*	mp-549706	P6 ₃ mc	0.1572	-1.2025	0.0812	0.0000
MgO	mp-1009127	Pm-3m	0.0813	-0.6164	0.2735	0.0000
MgS*	mp-1315	Fm-3m	0.1236	-0.7426	0.2438	0.0000
MgS	mp-13032	F-43m	0.2006	-1.1035	0.1624	0.0000
Na ₂ O*	mp-2352	Fm-3m	0.1191	-0.6209	0.0000	0.0000
Na ₂ O	mp-1975297	C2/m	0.1297	-0.6320	0.0003	0.0000
NaCl*	mp-22862	Fm-3m	0.0859	-0.5953	0.1111	0.0000
NaS*	mp-409	P-62m	0.4976	-2.6241	0.1126	0.0000
NaS	mp-1180106	C2/m	0.2104	-0.9839	0.0000	0.0000
PbSe*	mp-22009	Fmm2	0.4169	-1.6445	0.0576	0.0000
PbSe*	mp-2201	Fm-3m	0.3457	-1.4608	0.0807	0.0039
Rb ₂ O*	mp-1394	Fm-3m	0.0867	-1.2998	0.0368	0.0000
Rb ₂ O	mp-2023482	P-3m1	0.1302	-0.4067	0.0636	0.0000
Rb ₃ P	mp-974822	Fm-3m	0.1253	-0.5327	0.0666	0.0000
RbCl*	mp-23299	Pm-3m	0.0478	-0.6732	0.1019	0.0000
RbF*	mp-2064	Pm-3m	0.0879	-1.3700	0.0447	0.0036
RbF	mp-11718	Fm-3m	0.0460	-0.4536	0.0398	0.0000
RbI*	mp-22903	Fm-3m	0.0840	-0.4162	0.0534	0.0013
RbS	mp-1179684	C2/m	0.1226	-0.5861	0.0850	0.0092
Sc ₂ O ₃	mp-755313	R-3c	0.4210	-2.7698	0.0026	0.0000
SiO ₂ *	mp-9258	Pa-3	0.5130	-5.5969	0.0264	0.0000
SiO ₂	mp-10064	Fm-3m	0.3974	-3.7939	0.0240	0.0000
SnBr ₂	mp-978985	P4 ₂ /mnm	0.3595	-1.6332	0.1289	0.0000
SrF ₂ *	mp-1102240	Pnma	0.0589	-0.5401	0.2090	0.0000
SrF ₂	mp-1102911	Pnma	0.0505	-0.4858	0.2166	0.0000
SrS*	mp-10627	Pm-3m	0.0762	-0.4641	0.2376	0.0000
Te*	mp-19	P3 ₁ 21	0.3172	-1.1233	0.1197	0.0127
Te ₂ Ru*	mp-267	Pnnm	0.4723	-2.2556	0.0486	0.0017
TiO ₂ *	mp-430	P2 ₁ /c	0.5087	-3.0819	0.0145	0.0000
TiCl*	mp-571079	Cmcm	0.1717	-0.8409	0.1079	0.0000
Y ₂ O ₃ *	mp-558573	C2/m	0.4214	-2.9843	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §).

S5.3 Covalent Compounds

Table S10: Covalent compounds (non-metallic, $\text{ICOBI} \geq 0.5315$, observed range 0.5501 to 1.9431) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
As ₂ S ₃ *	mp-1189572	P-1	0.9136	-4.7416	0.0538	0.0000
As ₄ S ₃ *	mp-27543	Pnma	0.9714	-5.2731	0.0137	0.0000
As ₄ S ₃ *	mp-557321	Pnma	0.9717	-5.2816	0.0101	0.0000
AsF ₃ *	mp-28027	Pna2_1	0.6951	-6.3410	0.1114	0.0000
Au ₂ S*	mp-947	Pn-3m	0.6297	-4.8036	0.0181	0.0000
BI ₃ *	mp-23189	P6_3/m	1.2727	-6.8061	0.0232	0.0000
Be ₃ P ₂ *	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
Be ₃ P ₂ *	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
BeCl ₂ *	mp-23267	Ibam	0.7165	-5.6352	0.0021	0.0000
BeCl ₂	mp-1190080	I-43m	0.7528	-5.9613	0.0026	0.0000
Bi ₂ O ₃	mp-714859	P1	1.1110	-10.1326	0.0646	0.0001
Br*	mp-23154	Cmce	0.7827	-3.5362	0.4257	0.0097
C	mp-1080826	C2/m	0.9446	-9.5768	0.0000	0.0000
C	mp-1190171	Pnma	0.9445	-9.5798	0.0000	0.0000
C*	mp-66	Fd-3m	0.9497	-9.6602	0.0000	0.0000
C*	mp-47	P6_3/mmc	0.9482	-9.7832	0.0000	0.0000
C*	mp-169	R-3m	1.2173	-11.6891	0.0000	0.0000
C ₃ N ₄ *	mp-571653	P-43m	0.9531	-10.4956	0.0150	0.0000
C ₃ N ₄	mp-567885	P-6m2	1.2498	-13.1075	0.0143	0.0000
C ₃ N ₄ *	mp-570572	R3m	1.2498	-13.0884	0.0134	0.0000
C ₃ N ₄	mp-971683	P3_1	1.2479	-13.2995	0.0096	0.0000
C ₃ N ₄	mp-1182484	P6_3/m	1.9179	-15.3586	0.0000	0.0000
C ₃ N ₄	mp-2852	I-43d	0.9575	-10.9099	0.0106	0.0000
C ₃ N ₄	mp-563	Fd-3m	0.7047	-7.1477	0.0086	0.0000
Cl ₂ *	mp-22848	Cmce	0.9254	-5.5728	0.4555	0.0003
ClF ₃ *	mp-556767	Pnma	0.6390	-5.2173	0.1347	0.0000
F ₂ *	mp-561203	C2/c	0.9731	-6.9536	0.4569	0.0000
Ga ₂ Se ₃	mp-1224809	Imm2	0.8324	-4.7105	0.0085	0.0000
GeSe ₂ *	mp-10074	I-42d	0.9011	-4.9682	0.0171	0.0000
H ₂ *	mp-730101	P2_12_12_1	0.9954	-5.1222	0.0000	0.0000
H ₂ CN ₂ *	mp-703527	Pbca	1.9431	-16.8585	0.0000	0.0000
H ₃ N*	mp-643432	P2_12_12_1	0.8816	-8.0639	0.0000	0.0000
HC ₂ N ₃	mp-1103408	Cmc2_1	0.9561	-10.8107	0.0045	0.0000
HCl	mp-684609	Imm2	0.9480	-7.3985	0.0478	0.0001
IrCl ₃	mp-729507	C2	0.7112	-4.9995	0.0957	0.0000
Mn ₂ O ₇ *	mp-28338	P2_1/c	1.3609	-6.7339	0.0257	0.0000
MnN	mp-1009130	F-43m	0.6682	-3.9792	0.0247	0.0062
MoSe ₂ *	mp-1634	P6_3/mmc	0.6411	-2.8467	0.0035	0.0000
Nb ₂ O ₅ *	mp-1595	C2/m	0.7548	-4.9296	0.0075	0.0000
O ₂ *	mp-1524462	C2/m	1.4816	-17.7956	0.1271	0.2186
P*	mp-568348	P2/c	0.9202	-5.0760	0.0209	0.0000
P ₄ Se ₅ *	mp-2447	Pna2_1	0.9440	-5.0941	0.0348	0.0000
PS*	mp-612	C2/c	0.9840	-5.8992	0.0301	0.0000
PtCl ₂	mp-1219748	C2/m	0.7183	-5.3013	0.0756	0.0000
S*	mp-77	Fddd	0.9692	-5.9246	0.0880	0.0000
SN*	mp-236	P2_1/c	1.1480	-8.7413	0.0440	0.0000
Se*	mp-14	P3_121	0.9053	-4.1496	0.0828	0.0000
Si*	mp-149	Fd-3m	0.9243	-4.5418	0.0000	0.0000
SiO ₂ *	mp-546794	I-42d	0.7466	-7.9383	0.0071	0.0000
SiS ₂ *	mp-1095294	P2_1/c	0.9089	-5.7585	0.0044	0.0000
TeO ₂ *	mp-8377	P2_12_12_1	0.6215	-4.7653	0.0772	0.0000
TiO ₂ *	mp-1439	Pbcn	0.5838	-3.5796	0.0077	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
TiO ₂ *	mp-9173	Pnma	0.5897	-3.5580	0.0123	0.0000
ZrO ₂	mp-775909	P4 ₂ /mmn	0.5720	-4.1446	0.0017	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §).

S5.4 Mixed (Zintl-Type) Compounds

Table S11: Mixed-bonding (Zintl-type) compounds (a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide N₃⁻ ion ionically bound to its cation; observed range 0.8490 to 2.7322) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ba ₅ P ₄ *	mp-17566	Pnma	0.9040	-3.4927	0.2104	0.0000
BaP ₁₀ *	mp-28035	Cmc2 ₁	0.9058	-4.8297	0.0394	0.0000
BaS ₃ *	mp-556296	P2 ₁ 2 ₁ 2 ₁	0.9201	-5.2850	0.1084	0.0000
Bi ₂ O ₃	mp-674644	P1	0.9332	-7.5456	0.0131	0.0000
C(NO) ₂ *	mp-1197051	P2 ₁ 2 ₁ 2 ₁ 1	1.7617	-16.7609	0.0043	0.0000
C ₂ N ₂ O	mp-1214250	P6 ₃ /m	2.4576	-22.4067	0.0003	0.0000
CaN ₆	mp-656097	C2	1.8852	-17.7987	0.0004	0.0000
CaN ₆ *	mp-676	Fddd	1.8841	-18.1451	0.0003	0.0000
CdO ₂ *	mp-2310	Pa-3	0.9738	-7.1680	0.0540	0.0000
CdP ₄ *	mp-7904	P2 ₁ 1/c	0.9223	-5.1380	0.0094	0.0000
Cs ₂ As ₃ *	mp-15557	Fmmm	1.0431	-4.4248	0.0386	0.0000
CsN ₃ *	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN ₃	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsN ₃ *	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN ₃	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsP ₇ *	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP ₇ *	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.9336	-4.9318	0.0064	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.9336	-4.9318	0.0064	0.0000
H ₂ CN ₅	mp-1212563	C2/c	1.2926	-13.4719	0.0040	0.0000
H ₉ CN ₉ *	mp-1202455	P-1	1.2740	-12.5502	0.0042	0.0000
HC	mp-995197	C2/m	1.2662	-10.8409	0.0000	0.0000
HN	mp-1224501	P-1	0.8490	-8.0912	0.0231	0.0000
HN *	mp-23922	Pmna	1.8882	-19.1499	0.0000	0.0000
HN *	mp-720515	P2 ₁ 1/c	1.5928	-15.9550	0.0000	0.0000
KN ₃ *	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
KN ₃ *	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
KP ₁₅	mp-1211471	P-1	0.9260	-5.0473	0.0164	0.0000
LiN ₃ *	mp-2659	C2/m	1.8523	-20.2194	0.0000	0.0000
LiP ₅ *	mp-2412	Pna2 ₁	0.9037	-4.9376	0.0026	0.0000
LiP ₅ *	mp-2412	Pna2 ₁	0.9037	-4.9376	0.0026	0.0000
LiP ₅	mp-32760	Pna2 ₁	0.8771	-4.6811	0.0064	0.0000
LiP ₅	mp-32760	Pna2 ₁	0.8771	-4.6811	0.0064	0.0000
LiS ₄	mp-1143546	P1	0.9663	-5.6937	0.0304	0.0000
MgP ₄ *	mp-384	P2 ₁ 1/c	0.9196	-5.0216	0.0273	0.0000
N ₂ O *	mp-1180473	Pnma	2.3033	-22.3518	0.0013	0.0000
NaN ₃ *	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN ₃ *	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000
NaN ₃ *	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN ₃ *	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
NaN ₃	mp-634410	C2	1.8691	-20.2972	0.0000	0.0000
NaN ₃	mp-1065265	C2/m	2.7322	-23.2128	0.0006	0.0215
NaP ₁₅ *	mp-1194653	P-1	0.9227	-5.0620	0.0186	0.0000
NaP ₅ *	mp-31086	Pnma	0.9111	-4.9558	0.0285	0.0000
NaS*	mp-2400	P6 ₃ /mmc	0.9510	-4.7236	0.1279	0.0000
PI ₂ *	mp-29443	P-1	0.9586	-4.8005	0.1081	0.0000
PNO	mp-675742	P-1	1.1751	-12.3013	0.0406	0.0000
PbN ₆ *	mp-620058	Pna2 ₁	1.8686	-19.1852	0.0008	0.0000
PbN ₆ *	mp-667338	Pnma	1.8687	-19.2227	0.0007	0.0000
RbN ₃ *	mp-581833	P4/mmm	1.8969	-18.3403	0.0000	0.0000
RbN ₃	mp-1219567	Amm2	1.3756	-9.8572	0.0216	0.0000
RbP ₁₅	mp-1209298	P-1	0.9270	-5.0385	0.0156	0.0000
RbS*	mp-558071	Immm	0.9920	-5.0554	0.1964	0.0019
Si ₃ H	mp-995193	P-3m1	0.9370	-4.6600	0.0000	0.0000
SiH ₂ *	mp-1192532	P2 ₁ /m	0.9857	-4.7720	0.0000	0.0000
Sr ₃ As ₄	mp-678007	P1	0.8522	-3.0556	0.1161	0.0000
Sr ₃ P ₄ *	mp-14288	Fdd2	0.8910	-4.1353	0.1224	0.0000
Sr ₃ P ₄	mp-682953	P1	0.8889	-4.0705	0.1250	0.0000
SrN ₆ *	mp-2131	Fddd	1.8857	-18.4334	0.0000	0.0000
SrO ₂ *	mp-2697	I4/mmm	0.9835	-7.0573	0.1841	0.0000
SrO ₂ *	mp-726705	Pnma	0.9821	-6.8700	0.2389	0.0000
SrO ₂	mp-2763179	Pnma	0.9805	-6.9738	0.2327	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §).

S5.5 Unclassified Compounds

Table S12: Unclassified compounds (no ICOBI data available) — same columns, ICOBI/antibonding shown as “-” where missing.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
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Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §).

S6 LOBSTER Quality Control

Every LOBSTER run was screened using the fraction of k -points failing projection orthonormalization (`lobsterout`) and the maximum deviation reported in `bandOverlaps.lobster`, with a red-flag threshold of 0.1 on the latter (main text, Section 2.7). A systematic sweep of the full 597-structure dataset under the conservative combined criterion (100% k -point failure *and* band-overlap deviation > 0.5 *and* robust z -score > 10 on the resulting $\Delta(\text{ICOHP})$) flagged a single compound, a copper–gold intermetallic (Cu₄Au), for exclusion:

- 343/343 k -points fail orthonormalization (100%, not a borderline case);
- maximum `bandOverlaps.lobster` deviation of 11.4 (versus the 0.1 red-flag threshold);
- physically implausible ICOHP values (third-neighbor Cu–Cu contacts at 4.25–5.04 Å with ICOHP values of +7.6 to +11.6 eV, an order of magnitude too large to be physical);

- a $\Delta(\text{ICOHP})/\text{atom}$ of -380.5 eV, $\sim 27\times$ the next-worst case in the rest of the population.

The second-worst case identified by the sweep is already $29\times$ less extreme than Cu_4Au and is plausibly genuine, dense d -electron-rich intermetallic chemistry rather than a projection failure, confirming Cu_4Au as a unique quality-control case rather than a symptom of a wider problem. Every statistic reported in the main text has been computed on the remaining 588 (respectively 438–439 case-1-reaction) structures; the excluded compound’s own VASP+LOBSTER output is retained in the data repository for reference.

S7 Compute Cost

Table S13 summarizes VASP and LOBSTER wall-clock time across the 597 structures. Every job ran on a single SLURM node allocation (16 cores): VASP under MPI parallelism, then LOBSTER under OpenMP parallelism on the same 16 cores, sequentially within the same job.

Table S13: VASP and LOBSTER compute time (597 structures).

	VASP	LOBSTER
n	597	596 ^a
Mean (s)	188	578
Median (s)	72	157
Max (s)	5715 (1.6 h)	22555 (6.3 h)
Sequential total	31.2 h	95.8 h

^a one of 597 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 597 structures): 127.0 h — between two and three orders of magnitude below what a phonon (supercell-based) or AIMD screening of the same 597 structures, at the hierarchy described in the main text’s Introduction, would require, and consistent with the descriptor’s intended role as an inexpensive pre-screen rather than a replacement for that hierarchy.

S8 Case-1 Reaction Descriptor Values

Section S8.1 reproduces the project’s complete 517-reaction case-1 (decomposition-into-elements) population: for each compound, the balanced reaction (with each elemental reference’s space group), the resulting `ViabilityLabel` (`METASTABLE_VIABLE`/`UNSTABLE_NONEXISTENT`/insufficient data), and $\Delta(\text{ICOHP})/\text{atom}$ and $\Delta(\text{ICOHP}_{\text{antibonding}})/\text{atom}$ (main text, Eqs. 1–2). Sections S8.2 and S8.3 give the endobondic and exobondic subsets, respectively, for the 208-compound extension batch of the dataset (Section S1, Track 2), each reaction annotated with its E_{hull} . A condensed summary by stability family is given in Table S14.

S8.1 Complete Case-1 Reaction List

Table S14: Median $\Delta(\text{ICOHP})/\text{atom}$ and endobondic fraction by stability class ($n = 177$; the subset of the dataset carrying a three-way experimental-metastable/stable/theoretical-metastable family label, Section S1).

Class	n	Median $\Delta(\text{ICOHP})/\text{atom}$ (eV)	% endobondic
Experimental metastable	59	−1.038	28.8%
Stable	59	−1.391	22.0%
Theoretical metastable	59	−1.601	13.6%

Kruskal–Wallis across the three groups: $H = 2.56$, $p = 0.278$ (not significant at this subset size), but the ordering matches the expected physical trend: experimentally metastable compounds have the highest endobondic fraction, theoretical-only metastable compounds the lowest.

Table S15: The project’s complete 517-reaction case-1 (decomposition-to-elements) population (see §S8.2–§S8.3 for the same reactions restricted to the 208-compound extension campaign, split by sign) — compound (starred formula, mp-id, SG = space group), balanced reaction with each elemental reference’s space group in parentheses, viability label (`classify_viability()`: meta-viable = `METASTABLE_VIABLE`, not viable = `UNSTABLE_NONEXISTENT`, n/a = `insufficient_data`), $\Delta(\text{ICOHP})/\text{atom}$ in eV (§ of the main text) and $\Delta(\text{ICOHP anti-bonding})/\text{atom}$ in eV (§ of the main text, “−” where the full COHP trace is missing for the compound or one of its references).

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
Ag ₃ As	mp-1183193	P6 ₃ /mmc	Ag ₃ As → Ag (Fm-3m) + 0.3333 As (Cmce)	not viable	-6.057	0.001
Ag ₃ Pt*	mp-12065	Pm-3m	Ag ₃ Pt → Ag (Fm-3m) + 0.3333 Pt (Fm-3m)	stable	-9.557	0.010
Ag ₃ Sb*	mp-2273	Pmm2	Ag ₃ Sb → Ag (Fm-3m) + 0.3333 Sb (R-3m)	not viable	-4.798	0.003
AgH	mp-1096804	P6 ₃ mc	AgH → Ag (Fm-3m) + 0.5 H ₂ (P2 ₁₂ 12 ₁)	not viable	-3.003	0.000
AgI*	mp-22915	Pm-3m	AgI → Ag (Fm-3m) + I (Cmce)	stable	-2.317	0.012
AgN*	mp-1091417	Cmce	AgN → Ag (Fm-3m) + 0.5 N ₂ (−)	not viable	-5.126	−
AgN	mp-1009766	F-43m	AgN → Ag (Fm-3m) + 0.5 N ₂ (−)	not viable	-4.034	−
AgN	mp-1424734	Ama2	AgN → Ag (Fm-3m) + 0.5 N ₂ (−)	not viable	-5.629	−
AgN	mp-1428312	Pm-3m	AgN → Ag (Fm-3m) + 0.5 N ₂ (−)	not viable	-2.894	−
AgO*	mp-1079720	P2 ₁ /c	AgO → Ag (Fm-3m) + 0.5 O ₂ (C2/m)	stable	-4.662	0.009
AgPb ₃	mp-1183256	I4/mmm	AgPb ₃ → Ag (Fm-3m) + 3 Pb (Fm-3m)	not viable	-4.938	0.012
AgPd	mp-1183222	P-6m2	AgPd → Ag (Fm-3m) + Pd (Fm-3m)	stable	-1.460	0.010
Al ₁₂ Mo*	mp-550	Im-3	Al ₁₂ Mo → Al (Fm-3m) + 0.08333 Mo (Im-3m)	stable	-9.132	0.000
Al ₁₂ W*	mp-11227	Im-3	Al ₁₂ W → Al (Fm-3m) + 0.08333 W (Im-3m)	stable	-9.919	0.000
Al ₂ W*	mp-12524	P6 ₃ 22	Al ₂ W → Al (Fm-3m) + 0.5 W (Im-3m)	stable	-12.060	0.000

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
Al_3Ag	mp-1183130	Pm-3m	$\text{Al}_3\text{Ag} \rightarrow \text{Al (Fm-3m)} + 0.3333 \text{ Ag (Fm-3m)}$	not viable	-8.073	0.000
Al_3As	mp-1183129	Pm-3m	$\text{Al}_3\text{As} \rightarrow \text{Al (Fm-3m)} + 0.3333 \text{ As (Cmce)}$	not viable	-6.541	0.001
Al_3Os_2^*	mp-16521	I4/mmm	$\text{Al}_3\text{Os}_2 \rightarrow \text{Al (Fm-3m)} + 0.6667 \text{ Os (P6}_3\text{/mmc)}$	stable	-4.279	0.005
Al_4Si	mp-1228120	R-3m	$\text{Al}_4\text{Si} \rightarrow \text{Al (Fm-3m)} + 0.25 \text{ Si (Fd-3m)}$	meta-viable	0.660	-0.009
AlAg	mp-1228901	P-6m2	$\text{AlAg} \rightarrow \text{Al (Fm-3m)} + \text{Ag (Fm-3m)}$	stable	-4.553	0.000
AlAg_2	mp-985805	Cmcm	$\text{AlAg}_2 \rightarrow \text{Al (Fm-3m)} + 2 \text{ Ag (Fm-3m)}$	stable	-8.140	0.000
AlH_3^*	mp-1106356	Fd-3m	$\text{AlH}_3 \rightarrow \text{Al (Fm-3m)} + 1.5 \text{ H}_2 \text{ (P2}_1\text{2}_1\text{2}_1\text{)}$	stable	-2.280	0.000
AlNi^*	mp-1487	Pm-3m	$\text{AlNi} \rightarrow \text{Al (Fm-3m)} + \text{Ni (Fm-3m)}$	stable	3.020	0.015
AlSb	mp-1023918	Cmcm	$\text{AlSb} \rightarrow \text{Al (Fm-3m)} + \text{Sb (R-3m)}$	stable	-4.166	0.006
As_2S_3^*	mp-1189572	P-1	$\text{As}_2\text{S}_3 \rightarrow \text{As (Cmce)} + 1.5 \text{ S (Fddd)}$	stable	-0.306	0.003
As_3W_2^*	mp-11696	C2/m	$\text{As}_3\text{W}_2 \rightarrow \text{As (Cmce)} + 0.6667 \text{ W (Im-3m)}$	stable	-3.714	0.002
As_4S_3^*	mp-27543	Pnma	$\text{As}_4\text{S}_3 \rightarrow \text{As (Cmce)} + 0.75 \text{ S (Fddd)}$	stable	-0.167	0.009
As_4S_3^*	mp-557321	Pnma	$\text{As}_4\text{S}_3 \rightarrow \text{As (Cmce)} + 0.75 \text{ S (Fddd)}$	stable	-0.154	0.011
AsF_3^*	mp-28027	Pna2_1	$\text{AsF}_3 \rightarrow \text{As (Cmce)} + 1.5 \text{ F}_2 \text{ (C2/c)}$	stable	0.895	0.012
AsPd_2^*	mp-1080831	P-62m	$\text{AsPd}_2 \rightarrow \text{As (Cmce)} + 2 \text{ Pd (Fm-3m)}$	stable	-2.738	0.026
AsRh^*	mp-22079	Pnma	$\text{AsRh} \rightarrow \text{As (Cmce)} + \text{Rh (Fm-3m)}$	stable	-1.003	0.008
AsRh_2^*	mp-1102364	Pnma	$\text{AsRh}_2 \rightarrow \text{As (Cmce)} + 2 \text{ Rh (Fm-3m)}$	stable	-2.242	0.000
Au_2S^*	mp-947	Pn-3m	$\text{Au}_2\text{S} \rightarrow \text{Au (Fm-3m)} + 0.5 \text{ S (Fddd)}$	stable	-9.954	0.003
B_3Ir_2	mp-1228647	C2/m	$\text{B}_3\text{Ir}_2 \rightarrow \text{B (R-3m)} + 0.6667 \text{ Ir (Fm-3m)}$	stable	-6.439	0.006
B_6Pb	mp-1096825	Pm-3m	$\text{B}_6\text{Pb} \rightarrow \text{B (R-3m)} + 0.1667 \text{ Pb (Fm-3m)}$	not viable	-2.102	0.002
BI_3^*	mp-23189	P6_3/m	$\text{BI}_3 \rightarrow \text{B (R-3m)} + 3 \text{ I (Cmce)}$	stable	0.001	0.039
BPd_5	mp-1228665	I4/mmm	$\text{BPd}_5 \rightarrow \text{B (R-3m)} + 5 \text{ Pd (Fm-3m)}$	stable	-7.329	-0.009
BPd_5	mp-1228665	I4/mmm	$\text{BPd}_5 \rightarrow \text{B (R-3m)} + 5 \text{ Pd (Fm-3m)}$	stable	-7.324	-0.009
BW^*	mp-7832	I4_1/amd	$\text{BW} \rightarrow \text{B (R-3m)} + \text{W (Im-3m)}$	stable	-7.546	0.000
Ba_3Ca	mp-1183342	I4/mmm	$\text{Ba}_3\text{Ca} \rightarrow \text{Ba (Im-3m)} + 0.3333 \text{ Ca (Im-3m)}$	not viable	-0.210	-0.003
Ba_3N_2	mp-1013528	Pm-3m	$\text{Ba}_3\text{N}_2 \rightarrow \text{Ba (Im-3m)} + 0.3333 \text{ N}_2 \text{ (-)}$	not viable	-2.123	—
Ba_4Pd	mp-1214438	Fd-3m	$\text{Ba}_4\text{Pd} \rightarrow \text{Ba (Im-3m)} + 0.25 \text{ Pd (Fm-3m)}$	stable	-0.546	0.056
Ba_5P_4^*	mp-17566	Pnma	$\text{Ba}_5\text{P}_4 \rightarrow \text{Ba (Im-3m)} + 0.8 \text{ P (P2/c)}$	stable	-0.959	0.039
BaAu_2^*	mp-30363	P6/mmm	$\text{BaAu}_2 \rightarrow \text{Ba (Im-3m)} + 2 \text{ Au (Fm-3m)}$	stable	-8.739	0.045

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
BaCl_2^*	mp-568662	Fm-3m	$\text{BaCl}_2 \rightarrow \text{Ba (Im-3m)} + \text{Cl}_2$ (Cmce)	stable	-0.127	0.068
BaCl_2^*	mp-23199	Pnma	$\text{BaCl}_2 \rightarrow \text{Ba (Im-3m)} + \text{Cl}_2$ (Cmce)	stable	-0.224	0.067
BaF_3^*	mp-1080821	P2_1/m	$\text{BaF}_3 \rightarrow \text{Ba (Im-3m)} + 1.5 \text{ F}_2$ (C2/c)	stable	-0.868	0.047
BaF_3	mp-1183303	I4/mmm	$\text{BaF}_3 \rightarrow \text{Ba (Im-3m)} + 1.5 \text{ F}_2$ (C2/c)	stable	-0.249	0.049
BaH_2^*	mp-23715	Pnma	$\text{BaH}_2 \rightarrow \text{Ba (Im-3m)} + \text{H}_2$ (P2_12_12_1)	stable	-0.494	0.043
BaMg_3	mp-1016218	Pmm2	$\text{BaMg}_3 \rightarrow \text{Ba (Im-3m)} + 3 \text{ Mg}$ (P6_3/mmc)	meta-viable	0.072	-0.015
BaN_2^*	mp-1102371	Cc	$\text{BaN}_2 \rightarrow \text{Ba (Im-3m)} + \text{N}_2 (-)$	not viable	-5.342	—
BaO^*	mp-1008500	P6_3/mmc	$\text{BaO} \rightarrow \text{Ba (Im-3m)} + 0.5 \text{ O}_2$ (C2/m)	stable	-0.322	0.062
BaO	mp-1950989	P6_3/mmc	$\text{BaO} \rightarrow \text{Ba (Im-3m)} + 0.5 \text{ O}_2$ (C2/m)	stable	-1.613	0.063
BaP_{10}^*	mp-28035	Cmc2_1	$\text{BaP}_{10} \rightarrow \text{Ba (Im-3m)} + 10 \text{ P}$ (P2/c)	stable	-0.367	0.008
BaS_3^*	mp-556296	P2_12_12	$\text{BaS}_3 \rightarrow \text{Ba (Im-3m)} + 3 \text{ S}$ (Fddd)	stable	-0.623	0.036
BaSn_5^*	mp-571353	P6/mmm	$\text{BaSn}_5 \rightarrow \text{Ba (Im-3m)} + 5 \text{ Sn}$ (I4/mmm)	stable	-3.979	0.015
$\text{Be}_{12}\text{Mo}^*$	mp-1475	I4/mmm	$\text{Be}_{12}\text{Mo} \rightarrow \text{Be (P6_3/mmc)} + 0.08333 \text{ Mo (Im-3m)}$	stable	-2.103	0.000
$\text{Be}_{12}\text{Pd}^*$	mp-12666	I4/mmm	$\text{Be}_{12}\text{Pd} \rightarrow \text{Be (P6_3/mmc)} + 0.08333 \text{ Pd (Fm-3m)}$	stable	-2.118	0.003
Be_2Cu^*	mp-2031	Fd-3m	$\text{Be}_2\text{Cu} \rightarrow \text{Be (P6_3/mmc)} + 0.5 \text{ Cu (Fm-3m)}$	stable	-1.136	0.003
Be_2Nb_3^*	mp-11272	P4/mbm	$\text{Be}_2\text{Nb}_3 \rightarrow \text{Be (P6_3/mmc)} + 1.5 \text{ Nb (Im-3m)}$	stable	-5.616	0.000
Be_2V^*	mp-11281	P6_3/mmc	$\text{Be}_2\text{V} \rightarrow \text{Be (P6_3/mmc)} + 0.5 \text{ V (Im-3m)}$	stable	-3.002	0.000
Be_3N_2^*	mp-6977	P6_3/mmc	$\text{Be}_3\text{N}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.3333 \text{ N}_2 (-)$	stable	2.227	—
Be_3N_2^*	mp-18337	Ia-3	$\text{Be}_3\text{N}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.3333 \text{ N}_2 (-)$	stable	1.895	—
Be_3N_2	mp-1017550	P-3m1	$\text{Be}_3\text{N}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.3333 \text{ N}_2 (-)$	stable	2.366	—
Be_3P_2^*	mp-567841	Ia-3	$\text{Be}_3\text{P}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.6667 \text{ P (P2/c)}$	stable	-0.211	0.004
Be_3P_2	mp-1084771	Pn-3m	$\text{Be}_3\text{P}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.6667 \text{ P (P2/c)}$	stable	-0.180	0.004
Be_3P_2^*	mp-567841	Ia-3	$\text{Be}_3\text{P}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.6667 \text{ P (P2/c)}$	stable	-0.211	0.004
Be_3P_2	mp-1084771	Pn-3m	$\text{Be}_3\text{P}_2 \rightarrow \text{Be (P6_3/mmc)} + 0.6667 \text{ P (P2/c)}$	stable	-0.180	0.004
BeAu_3	mp-983539	Pm-3m	$\text{BeAu}_3 \rightarrow \text{Be (P6_3/mmc)} + 3 \text{ Au (Fm-3m)}$	not viable	-11.262	0.000
BeCl_2^*	mp-23267	Ibam	$\text{BeCl}_2 \rightarrow \text{Be (P6_3/mmc)} + \text{Cl}_2$ (Cmce)	stable	1.820	0.050
BeCl_2	mp-1172909	I-43m	$\text{BeCl}_2 \rightarrow \text{Be (P6_3/mmc)} + \text{Cl}_2$ (Cmce)	not viable	-5.781	-0.714
BeCl_2	mp-1190080	I-43m	$\text{BeCl}_2 \rightarrow \text{Be (P6_3/mmc)} + \text{Cl}_2$ (Cmce)	stable	1.987	0.050

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHP}_{\text{ab.}}$
BeCu [*]	mp-2323	Pm-3m	BeCu $\rightarrow$ Be (P6_3/mmc) + Cu (Fm-3m)	stable	5.037	0.004
BeF ₂ [*]	mp-15951	P3_121	BeF ₂ $\rightarrow$ Be (P6_3/mmc) + F ₂ (C2/c)	stable	3.151	0.027
BeF ₂	mp-975644	P6/mmm	BeF ₂ $\rightarrow$ Be (P6_3/mmc) + F ₂ (C2/c)	stable	-0.682	0.019
BeO [*]	mp-7599	P4_2/mnm	BeO $\rightarrow$ Be (P6_3/mmc) + 0.5 O ₂ (C2/m)	stable	2.786	0.015
BeO	mp-1778	F-43m	BeO $\rightarrow$ Be (P6_3/mmc) + 0.5 O ₂ (C2/m)	stable	4.439	0.015
C(NO) ₂ [*]	mp-1197051	P2_12_12_1	C(NO) ₂ $\rightarrow$ C (P6_3/mmc) + N ₂ (-) + O ₂ (C2/m)	meta-viable	2.172	—
C ₂ N ₂ O	mp-1214250	P6_3/m	C ₂ N ₂ O $\rightarrow$ C (P6_3/mmc) + 0.5 N ₂ (-) + 0.25 O ₂ (C2/m)	stable	1.361	—
C ₃ N ₄ [*]	mp-571653	P-43m	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	3.934	—
C ₃ N ₄	mp-567885	P-6m2	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	2.903	—
C ₃ N ₄ [*]	mp-570572	R3m	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	2.886	—
C ₃ N ₄	mp-971683	P3_1	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	3.229	—
C ₃ N ₄	mp-1182484	P6_3/m	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	not viable	-0.826	—
C ₃ N ₄	mp-2852	I-43d	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	4.659	—
C ₃ N ₄	mp-563	Fd-3m	C ₃ N ₄ $\rightarrow$ C (P6_3/mmc) + 0.6667 N ₂ (-)	meta-viable	2.364	—
Ca ₃ N ₂ [*]	mp-1047	R-3c	Ca ₃ N ₂ $\rightarrow$ Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.519	—
Ca ₃ N ₂ [*]	mp-844	Ia-3	Ca ₃ N ₂ $\rightarrow$ Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.468	—
Ca ₃ N ₂	mp-1013524	Pm-3m	Ca ₃ N ₂ $\rightarrow$ Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.520	—
Ca ₃ Zn	mp-1039229	I4/mmm	Ca ₃ Zn $\rightarrow$ Ca (Im-3m) + 0.3333 Zn (P6_3/mmc)	not viable	-3.115	0.023
CaAl ₄ [*]	mp-1749	I4/mmm	CaAl ₄ $\rightarrow$ Ca (Im-3m) + 4 Al (Fm-3m)	stable	-8.981	0.011
CaCl ₂ [*]	mp-23214	Pnnm	CaCl ₂ $\rightarrow$ Ca (Im-3m) + Cl ₂ (Cmce)	stable	-0.320	0.051
CaCl ₂	mp-1120739	Fm-3m	CaCl ₂ $\rightarrow$ Ca (Im-3m) + Cl ₂ (Cmce)	stable	-0.221	0.051
CaF ₂ [*]	mp-10464	Pnma	CaF ₂ $\rightarrow$ Ca (Im-3m) + F ₂ (C2/c)	stable	-0.223	0.036
CaF ₂	mp-554355	P4/mmm	CaF ₂ $\rightarrow$ Ca (Im-3m) + F ₂ (C2/c)	stable	-0.181	0.042
CaGa ₂ [*]	mp-992	P6/mmm	CaGa ₂ $\rightarrow$ Ca (Im-3m) + 2 Ga (Cmce)	stable	-1.011	0.016
CaGa ₄ [*]	mp-2461	C2/m	CaGa ₄ $\rightarrow$ Ca (Im-3m) + 4 Ga (Cmce)	stable	-1.077	0.011
CaGe ₂	mp-643542	C2/m	CaGe ₂ $\rightarrow$ Ca (Im-3m) + 2 Ge (Fd-3m)	stable	-1.451	0.016
CaN [*]	mp-1058549	Fm-3m	CaN $\rightarrow$ Ca (Im-3m) + 0.5 N ₂ (-)	stable	-2.456	—
CaN ₆	mp-656097	C2	CaN ₆ $\rightarrow$ Ca (Im-3m) + 3 N ₂ (-)	stable	1.449	—
CaN ₆ [*]	mp-676	Fddd	CaN ₆ $\rightarrow$ Ca (Im-3m) + 3 N ₂ (-)	stable	1.698	—

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
CaNi*	mp-1059707	Pm-3m	CaNi $\rightarrow$ Ca (Im-3m) + Ni (Fm-3m)	not viable	-1.925	0.031
CaO*	mp-1079707	Cmce	CaO $\rightarrow$ Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-2.213	0.030
CaO*	mp-2605	Fm-3m	CaO $\rightarrow$ Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-0.952	0.043
CaO	mp-1181903	Fd-3m	CaO $\rightarrow$ Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-2.997	0.039
CaSn*	mp-2450	Cmcm	CaSn $\rightarrow$ Ca (Im-3m) + Sn (I4/mmm)	stable	-2.557	0.026
CdI ₂ *	mp-567259	P-3m1	CdI ₂ $\rightarrow$ Cd (P6 ₃ /mmc) + 2 I (Cmce)	stable	-0.226	0.021
CdIn ₃	mp-1183674	Fm-3m	CdIn ₃ $\rightarrow$ Cd (P6 ₃ /mmc) + 3 In (I4/mmm)	not viable	-6.877	0.000
CdO ₂ *	mp-2310	Pa-3	CdO ₂ $\rightarrow$ Cd (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-2.628	0.008
CdP ₄ *	mp-7904	P2 ₁ /c	CdP ₄ $\rightarrow$ Cd (P6 ₃ /mmc) + 4 P (P2/c)	stable	-0.512	0.004
CdPd*	mp-1696	P4/mmm	CdPd $\rightarrow$ Cd (P6 ₃ /mmc) + Pd (Fm-3m)	stable	-6.170	0.021
CdSb ₃	mp-1183739	P6 ₃ /mmc	CdSb ₃ $\rightarrow$ Cd (P6 ₃ /mmc) + 3 Sb (R-3m)	not viable	-1.756	-0.002
ClF ₃ *	mp-556767	Pnma	ClF ₃ $\rightarrow$ Cl ₂ (Cmce) + 3 F ₂ (C2/c)	stable	0.688	0.032
CoB*	mp-20857	Pnma	CoB $\rightarrow$ Co (Fm-3m) + B (R-3m)	stable	-4.753	0.001
CoGe*	mp-21237	C2/m	CoGe $\rightarrow$ Co (Fm-3m) + Ge (Fd-3m)	stable	-2.962	0.003
CoRe	mp-1225995	P-6m2	CoRe $\rightarrow$ Co (Fm-3m) + Re (P6 ₃ /mmc)	not viable	-1.039	0.003
CoSe ₂	mp-1390196	Fd-3m	CoSe ₂ $\rightarrow$ Co (Fm-3m) + 2 Se (P3 ₁₂₁)	stable	-0.631	0.003
CoSn*	mp-20536	P6/mmm	CoSn $\rightarrow$ Co (Fm-3m) + Sn (I4/mmm)	stable	-5.095	0.005
Cr ₂ C	mp-1226378	P-3m1	Cr ₂ C $\rightarrow$ Cr (Im-3m) + 0.5 C (P6 ₃ /mmc)	stable	-0.259	-0.000
Cr ₅ Si ₃ *	mp-7506	I4/mcm	Cr ₅ Si ₃ $\rightarrow$ Cr (Im-3m) + 0.6 Si (Fd-3m)	stable	0.725	-0.000
CrMo	mp-1226200	Cmmm	CrMo $\rightarrow$ Cr (Im-3m) + Mo (Im-3m)	not viable	-1.340	0.001
CrRe ₃	mp-1183703	Pm-3m	CrRe ₃ $\rightarrow$ Cr (Im-3m) + 3 Re (P6 ₃ /mmc)	not viable	-3.432	0.004
CrW	mp-1226162	Cmmm	CrW $\rightarrow$ Cr (Im-3m) + W (Im-3m)	not viable	-4.060	0.001
Cs ₂ As ₃ *	mp-15557	Fmmm	Cs ₂ As ₃ $\rightarrow$ Cs (Im-3m) + 1.5 As (Cmce)	stable	-0.139	-0.004
Cs ₂ S*	mp-1077814	P6 ₃ /mmc	Cs ₂ S $\rightarrow$ Cs (Im-3m) + 0.5 S (Fddd)	stable	0.955	0.007
Cs ₂ Se*	mp-1011696	Pnma	Cs ₂ Se $\rightarrow$ Cs (Im-3m) + 0.5 Se (P3 ₁₂₁)	stable	0.673	-0.004
CsCl*	mp-573697	Fm-3m	CsCl $\rightarrow$ Cs (Im-3m) + 0.5 Cl ₂ (Cmce)	stable	0.883	0.039
CsF*	mp-8455	Pm-3m	CsF $\rightarrow$ Cs (Im-3m) + 0.5 F ₂ (C2/c)	stable	3.846	0.016
CsH*	mp-632319	Pm-3m	CsH $\rightarrow$ Cs (Im-3m) + 0.5 H ₂ (P2 _{12_12_1})	stable	3.123	-0.005

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
CsN_3^*	mp-510557	I4/mcm	$\text{CsN}_3 \rightarrow \text{Cs (Im-3m)} + 1.5 \text{ N}_2 (-)$	stable	1.402	–
CsN_3	mp-1225892	P4/mmm	$\text{CsN}_3 \rightarrow \text{Cs (Im-3m)} + 1.5 \text{ N}_2 (-)$	stable	2.026	–
CsN_3^*	mp-510557	I4/mcm	$\text{CsN}_3 \rightarrow \text{Cs (Im-3m)} + 1.5 \text{ N}_2 (-)$	stable	1.402	–
CsN_3	mp-1225892	P4/mmm	$\text{CsN}_3 \rightarrow \text{Cs (Im-3m)} + 1.5 \text{ N}_2 (-)$	stable	2.026	–
CsO_2^*	mp-1441	I4/mmm	$\text{CsO}_2 \rightarrow \text{Cs (Im-3m)} + \text{O}_2$ (C2/m)	stable	0.067	-0.004
CsO_2	mp-1096936	Cmmm	$\text{CsO}_2 \rightarrow \text{Cs (Im-3m)} + \text{O}_2$ (C2/m)	stable	0.774	-0.003
CsP_7^*	mp-1201790	Pbcm	$\text{CsP}_7 \rightarrow \text{Cs (Im-3m)} + 7 \text{ P}$ (P2/c)	stable	-0.298	0.008
CsP_7^*	mp-1201790	Pbcm	$\text{CsP}_7 \rightarrow \text{Cs (Im-3m)} + 7 \text{ P}$ (P2/c)	stable	-0.298	0.008
CsP_7	mp-1213206	Pca2_1	$\text{CsP}_7 \rightarrow \text{Cs (Im-3m)} + 7 \text{ P}$ (P2/c)	stable	-0.267	0.008
CsP_7	mp-1213206	Pca2_1	$\text{CsP}_7 \rightarrow \text{Cs (Im-3m)} + 7 \text{ P}$ (P2/c)	stable	-0.267	0.008
CsRb	mp-1225895	Cmmm	$\text{CsRb} \rightarrow \text{Cs (Im-3m)} + \text{Rb (Im-3m)}$	not viable	-0.005	-0.001
Cu_2S^*	mp-12087	Fm-3m	$\text{Cu}_2\text{S} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ S}$ (Fddd)	stable	-1.155	-0.000
Cu_2S^*	mp-553942	P6_3/mmc	$\text{Cu}_2\text{S} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ S}$ (Fddd)	not viable	-1.335	0.003
Cu_3Ge^*	mp-19724	Pmmn	$\text{Cu}_3\text{Ge} \rightarrow \text{Cu (Fm-3m)} + 0.3333 \text{ Ge (Fd-3m)}$	not viable	-1.198	-0.004
Cu_3Te_2	mp-1225676	P4mm	$\text{Cu}_3\text{Te}_2 \rightarrow \text{Cu (Fm-3m)} + 0.6667 \text{ Te (P3_121)}$	stable	-0.455	-0.006
Cu_7Te_4^*	mp-624307	P3m1	$\text{Cu}_7\text{Te}_4 \rightarrow \text{Cu (Fm-3m)} + 0.5714 \text{ Te (P3_121)}$	not viable	-0.833	-0.003
CuH	mp-1225705	P-3m1	$\text{CuH} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ H}_2$ (P2_12_12_1)	not viable	-0.444	-0.000
CuNi	mp-1184069	P-6m2	$\text{CuNi} \rightarrow \text{Cu (Fm-3m)} + \text{Ni (Fm-3m)}$	not viable	-1.223	-0.009
FeS^*	mp-505531	P4/nmm	$\text{FeS} \rightarrow \text{Fe (Im-3m)} + \text{S (Fddd)}$	stable	0.183	0.006
FeTe_2^*	mp-1102002	Pa-3	$\text{FeTe}_2 \rightarrow \text{Fe (Im-3m)} + 2 \text{ Te}$ (P3_121)	stable	0.919	0.012
Ga_2Au_7^*	mp-571598	P-62m	$\text{Ga}_2\text{Au}_7 \rightarrow \text{Ga (Cmce)} + 3.5 \text{ Au}$ (Fm-3m)	stable	-11.064	0.000
Ga_2Se_3	mp-1224809	Imm2	$\text{Ga}_2\text{Se}_3 \rightarrow \text{Ga (Cmce)} + 1.5 \text{ Se}$ (P3_121)	stable	1.407	0.009
Ga_3Hg	mp-1184289	P6_3/mmc	$\text{Ga}_3\text{Hg} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Hg (P6/mmm)}$	not viable	-1.450	0.000
Ga_3Pb	mp-1184425	I4/mmm	$\text{Ga}_3\text{Pb} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Pb (Fm-3m)}$	not viable	-2.802	0.003
Ga_3Re^*	mp-1188393	P4_2/mnm	$\text{Ga}_3\text{Re} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Re (P6_3/mmc)}$	stable	-1.278	0.002
GaSb	mp-1224785	I-4m2	$\text{GaSb} \rightarrow \text{Ga (Cmce)} + \text{Sb (R-3m)}$	stable	1.090	0.005
Ge_2Os^*	mp-10032	C2/m	$\text{Ge}_2\text{Os} \rightarrow \text{Ge (Fd-3m)} + 0.5 \text{ Os}$ (P6_3/mmc)	stable	0.178	0.003
Ge_3Mo_5^*	mp-17094	I4/mcm	$\text{Ge}_3\text{Mo}_5 \rightarrow \text{Ge (Fd-3m)} + 1.667 \text{ Mo (Im-3m)}$	stable	-4.378	0.001

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHP}_{\text{ab.}}$
Ge ₃ O	mp-1184665	I4/mmm	Ge ₃ O $\rightarrow$ Ge (Fd-3m) + 0.1667 O ₂ (C2/m)	not viable	-2.292	0.006
Ge ₄ Rh*	mp-1104286	P3_121	Ge ₄ Rh $\rightarrow$ Ge (Fd-3m) + 0.25 Rh (Fm-3m)	stable	-1.092	0.006
GeAs*	mp-7591	I4mm	GeAs $\rightarrow$ Ge (Fd-3m) + As (Cmce)	meta-viable	0.008	-0.000
GeS	mp-12910	Cmcm	GeS $\rightarrow$ Ge (Fd-3m) + S (Fddd)	stable	-1.256	0.004
GeSb	mp-9935	Fm-3m	GeSb $\rightarrow$ Ge (Fd-3m) + Sb (R-3m)	not viable	-0.211	0.003
GeSe ₂ *	mp-10074	I-42d	GeSe ₂ $\rightarrow$ Ge (Fd-3m) + 2 Se (P3_121)	stable	0.772	0.007
GeTe*	mp-938	R3m	GeTe $\rightarrow$ Ge (Fd-3m) + Te (P3_121)	stable	-1.063	-0.007
H ₂ CN ₂ *	mp-703527	Pbca	H ₂ CN ₂ $\rightarrow$ H ₂ (P2_12_12_1) + C (P6_3/mmc) + N ₂ (-)	stable	1.140	-
H ₂ CN ₅	mp-1212563	C2/c	H ₂ CN ₅ $\rightarrow$ H ₂ (P2_12_12_1) + C (P6_3/mmc) + 2.5 N ₂ (-)	stable	3.057	-
H ₂ O*	mp-2739191	C222_1	H ₂ O $\rightarrow$ H ₂ (P2_12_12_1) + 0.5 O ₂ (C2/m)	stable	1.692	0.009
H ₃ N*	mp-643432	P2_12_12_1	H ₃ N $\rightarrow$ H ₂ (P2_12_12_1) + 0.3333 N ₂ (-)	stable	1.582	-
H ₃ Pd ₄	mp-1224413	R-3m	H ₃ Pd ₄ $\rightarrow$ H ₂ (P2_12_12_1) + 2.667 Pd (Fm-3m)	stable	-2.188	0.006
H ₄ N	mp-1238814	P-43m	H ₄ N $\rightarrow$ H ₂ (P2_12_12_1) + 0.25 N ₂ (-)	meta-viable	0.973	-
H ₉ CN ₉ *	mp-1202455	P-1	H ₉ CN ₉ $\rightarrow$ H ₂ (P2_12_12_1) + 0.2222 C (P6_3/mmc) + N ₂ (-)	stable	2.117	-
HC	mp-995197	C2/m	HC $\rightarrow$ H ₂ (P2_12_12_1) + 2 C (P6_3/mmc)	stable	-0.470	0.000
HC ₂	mp-995184	C2/m	HC ₂ $\rightarrow$ H ₂ (P2_12_12_1) + 4 C (P6_3/mmc)	not viable	-0.680	0.000
HC ₂ N ₃	mp-1103408	Cmc2_1	HC ₂ N ₃ $\rightarrow$ H ₂ (P2_12_12_1) + 4 C (P6_3/mmc) + 3 N ₂ (-)	meta-viable	4.081	-
HC ₃	mp-995200	P2_1/m	HC ₃ $\rightarrow$ H ₂ (P2_12_12_1) + 6 C (P6_3/mmc)	not viable	-0.400	0.000
HC ₆	mp-995202	P-1	HC ₆ $\rightarrow$ H ₂ (P2_12_12_1) + 12 C (P6_3/mmc)	not viable	-0.350	0.000
HCl	mp-684609	Imm2	HCl $\rightarrow$ H ₂ (P2_12_12_1) + Cl ₂ (Cmce)	stable	0.990	0.030
HI*	mp-697025	C2/c	HI $\rightarrow$ H ₂ (P2_12_12_1) + 2 I (Cmce)	stable	0.675	0.025
HN	mp-1224501	P-1	HN $\rightarrow$ H ₂ (P2_12_12_1) + N ₂ (-)	stable	2.269	-
HN*	mp-23922	Pmna	HN $\rightarrow$ H ₂ (P2_12_12_1) + N ₂ (-)	stable	2.549	-
HN*	mp-720515	P2_1/c	HN $\rightarrow$ H ₂ (P2_12_12_1) + N ₂ (-)	stable	2.339	-
Hf ₃ Pb	mp-1184772	I4/mmm	Hf ₃ Pb $\rightarrow$ Hf (P6_3/mmc) + 0.3333 Pb (Fm-3m)	stable	-2.735	0.004
HfCd*	mp-1007758	P4/nmm	HfCd $\rightarrow$ Hf (P6_3/mmc) + Cd (P6_3/mmc)	stable	-0.904	0.000
HfCr ₂ *	mp-569624	P6_3/mmc	HfCr ₂ $\rightarrow$ Hf (P6_3/mmc) + 2 Cr (Im-3m)	not viable	-1.299	0.001
HfCu ₃	mp-1184704	I4/mmm	HfCu ₃ $\rightarrow$ Hf (P6_3/mmc) + 3 Cu (Fm-3m)	stable	-1.577	0.004

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
HfMg ₁₅	mp-1023478	P-6m2	HfMg ₁₅ $\rightarrow$ Hf (P6_3/mmc) + 15 Mg (P6_3/mmc)	not viable	-0.443	-0.012
HfMo	mp-1224314	Cmmm	HfMo $\rightarrow$ Hf (P6_3/mmc) + Mo (Im-3m)	not viable	-2.315	0.002
HfTc ₂ *	mp-1095669	P6_3/mmc	HfTc ₂ $\rightarrow$ Hf (P6_3/mmc) + 2 Tc (P6_3/mmc)	stable	-3.592	0.002
HfW ₂	mp-1224230	Fmmm	HfW ₂ $\rightarrow$ Hf (P6_3/mmc) + 2 W (Im-3m)	not viable	-7.285	0.000
Hg ₃ Pb	mp-1184700	P6_3/mmc	Hg ₃ Pb $\rightarrow$ Hg (P6/mmm) + 0.3333 Pb (Fm-3m)	not viable	-1.457	0.004
Hg ₃ Sb	mp-977242	P6_3/mmc	Hg ₃ Sb $\rightarrow$ Hg (P6/mmm) + 0.3333 Sb (R-3m)	not viable	-0.580	0.003
Hg ₄ Pt*	mp-936	Im-3m	Hg ₄ Pt $\rightarrow$ Hg (P6/mmm) + 0.25 Pt (Fm-3m)	stable	-2.591	0.009
Hg ₅ Pd ₂ *	mp-30721	P4/mbm	Hg ₅ Pd ₂ $\rightarrow$ Hg (P6/mmm) + 0.4 Pd (Fm-3m)	stable	0.011	0.012
HgF ₂ *	mp-8177	Fm-3m	HgF ₂ $\rightarrow$ Hg (P6/mmm) + F ₂ (C2/c)	stable	0.984	-0.004
IF ₇ *	mp-27988	Aea2	IF ₇ $\rightarrow$ I (Cmce) + 3.5 F ₂ (C2/c)	stable	1.714	0.036
IN ₂	mp-1097011	I4/mcm	IN ₂ $\rightarrow$ I (Cmce) + N ₂ (-)	stable	-0.079	-
In ₂ Te ₃ *	mp-1105025	I4mm	In ₂ Te ₃ $\rightarrow$ In (I4/mmm) + 1.5 Te (P3_121)	stable	-2.367	0.003
In ₃ Ir*	mp-630976	P4_2/mnm	In ₃ Ir $\rightarrow$ In (I4/mmm) + 0.3333 Ir (Fm-3m)	stable	-8.031	0.005
In ₇ Cu ₁₀ *	mp-646039	C2/m	In ₇ Cu ₁₀ $\rightarrow$ In (I4/mmm) + 1.429 Cu (Fm-3m)	stable	-3.068	0.000
InBr*	mp-22870	Cmcm	InBr $\rightarrow$ In (I4/mmm) + Br (Cmce)	stable	-3.996	0.002
InCl*	mp-571636	Cmc2_1	InCl $\rightarrow$ In (I4/mmm) + 0.5 Cl ₂ (Cmce)	stable	-4.587	0.012
InNi*	mp-20997	Pm-3m	InNi $\rightarrow$ In (I4/mmm) + Ni (Fm-3m)	stable	-0.224	0.014
InSe ₂	mp-1232036	P4/nmm	InSe ₂ $\rightarrow$ In (I4/mmm) + 2 Se (P3_121)	stable	-1.441	0.004
InSn ₄	mp-1223653	Cmmm	InSn ₄ $\rightarrow$ In (I4/mmm) + 4 Sn (I4/mmm)	not viable	-0.425	0.006
Ir ₃ Ru	mp-974358	I4/mmm	Ir ₃ Ru $\rightarrow$ Ir (Fm-3m) + 0.3333 Ru (P6_3/mmc)	stable	-11.896	0.006
Ir ₄ W	mp-1223666	R-3m	Ir ₄ W $\rightarrow$ Ir (Fm-3m) + 0.25 W (Im-3m)	stable	-14.283	-0.034
IrCl ₃	mp-729507	C2	IrCl ₃ $\rightarrow$ Ir (Fm-3m) + 1.5 Cl ₂ (Cmce)	stable	-1.896	0.015
IrPd	mp-1223675	R-3m	IrPd $\rightarrow$ Ir (Fm-3m) + Pd (Fm-3m)	not viable	-6.182	-0.004
K ₂ O*	mp-971	Fm-3m	K ₂ O $\rightarrow$ K (Fm-3m) + 0.25 O ₂ (C2/m)	stable	-0.436	0.014
K ₂ O	mp-1223784	P4/mmm	K ₂ O $\rightarrow$ K (Fm-3m) + 0.25 O ₂ (C2/m)	stable	-1.564	0.006
K ₂ S*	mp-559278	P6_3/mmc	K ₂ S $\rightarrow$ K (Fm-3m) + 0.5 S (Fddd)	stable	-0.229	0.014
K ₂ S*	mp-1022	Fm-3m	K ₂ S $\rightarrow$ K (Fm-3m) + 0.5 S (Fddd)	stable	-0.336	0.016
K ₃ Rb	mp-976115	P6_3/mmc	K ₃ Rb $\rightarrow$ K (Fm-3m) + 0.3333 Rb (Im-3m)	not viable	-0.377	-0.001
K ₅ As ₄	mp-684799	P-1	K ₅ As ₄ $\rightarrow$ K (Fm-3m) + 0.8 As (Cmce)	stable	-0.492	-0.006

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
KCl*	mp-23289	Pm-3m	KCl $\rightarrow$ K (Fm-3m) + 0.5 Cl ₂ (Cmce)	stable	0.762	0.025
KF ₃ *	mp-1544557	P2_12_12_1	KF ₃ $\rightarrow$ K (Fm-3m) + 1.5 F ₂ (C2/c)	stable	-1.151	0.009
KF ₃	mp-2749823	P2_1	KF ₃ $\rightarrow$ K (Fm-3m) + 1.5 F ₂ (C2/c)	stable	-1.151	0.009
KI*	mp-22898	Fm-3m	KI $\rightarrow$ K (Fm-3m) + I (Cmce)	stable	0.550	0.030
KMg ₁₅	mp-1023498	P-6m2	KMg ₁₅ $\rightarrow$ K (Fm-3m) + 15 Mg (P6_3/mmc)	meta-viable	0.008	-0.016
KN ₂ *	mp-1071868	Cmc2_1	KN ₂ $\rightarrow$ K (Fm-3m) + N ₂ (-)	not viable	-0.142	-
KN ₂	mp-1180757	Cmcm	KN ₂ $\rightarrow$ K (Fm-3m) + N ₂ (-)	not viable	-1.004	-
KN ₃ *	mp-827	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	stable	1.821	-
KN ₃ *	mp-827	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	stable	1.821	-
KN ₃	mp-636056	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	not viable	-2.306	-
KP*	mp-1058315	R-3m	KP $\rightarrow$ K (Fm-3m) + P (P2/c)	not viable	-2.160	0.006
KP	mp-1057787	Immm	KP $\rightarrow$ K (Fm-3m) + P (P2/c)	not viable	-2.137	0.006
KP ₁₅	mp-1211471	P-1	KP ₁₅ $\rightarrow$ K (Fm-3m) + 15 P (P2/c)	stable	-0.116	0.002
KSn ₃	mp-1185151	P6_3/mmc	KSn ₃ $\rightarrow$ K (Fm-3m) + 3 Sn (I4/mmm)	stable	-3.780	0.000
Li ₁₃ Sn ₅ *	mp-30769	P-3m1	Li ₁₃ Sn ₅ $\rightarrow$ Li (R-3m) + 0.3846 Sn (I4/mmm)	stable	-0.212	0.002
Li ₂ O*	mp-1960	Fm-3m	Li ₂ O $\rightarrow$ Li (R-3m) + 0.25 O ₂ (C2/m)	stable	2.337	0.010
Li ₂ O	mp-755894	Pnma	Li ₂ O $\rightarrow$ Li (R-3m) + 0.25 O ₂ (C2/m)	stable	1.948	0.010
Li ₂ S*	mp-1125	Pnma	Li ₂ S $\rightarrow$ Li (R-3m) + 0.5 S (Fddd)	stable	1.850	0.011
Li ₂ S	mp-557142	P6_3/mmc	Li ₂ S $\rightarrow$ Li (R-3m) + 0.5 S (Fddd)	stable	1.505	0.011
Li ₂ Se*	mp-2286	Fm-3m	Li ₂ Se $\rightarrow$ Li (R-3m) + 0.5 Se (P3_121)	stable	2.422	0.007
Li ₃ Ag	mp-976408	I4/mmm	Li ₃ Ag $\rightarrow$ Li (R-3m) + 0.3333 Ag (Fm-3m)	stable	-0.653	0.000
Li ₃ Ca	mp-976075	I4/mmm	Li ₃ Ca $\rightarrow$ Li (R-3m) + 0.3333 Ca (Im-3m)	not viable	-0.886	0.016
Li ₃ Hg*	mp-1646	Fm-3m	Li ₃ Hg $\rightarrow$ Li (R-3m) + 0.3333 Hg (P6/mmm)	stable	0.598	0.000
Li ₃ N*	mp-2341	P6_3/mmc	Li ₃ N $\rightarrow$ Li (R-3m) + 0.1667 N ₂ (-)	stable	1.847	-
LiB*	mp-1001835	P6_3/mmc	LiB $\rightarrow$ Li (R-3m) + B (R-3m)	stable	-0.905	0.000
LiBr*	mp-23259	Fm-3m	LiBr $\rightarrow$ Li (R-3m) + Br (Cmce)	stable	4.058	0.030
LiCl*	mp-22905	Fm-3m	LiCl $\rightarrow$ Li (R-3m) + 0.5 Cl ₂ (Cmce)	stable	2.803	0.038
LiCl	mp-1185319	P6_3mc	LiCl $\rightarrow$ Li (R-3m) + 0.5 Cl ₂ (Cmce)	stable	2.500	0.038
LiF*	mp-1138	Fm-3m	LiF $\rightarrow$ Li (R-3m) + 0.5 F ₂ (C2/c)	stable	4.467	0.021
LiF	mp-1009009	Pm-3m	LiF $\rightarrow$ Li (R-3m) + 0.5 F ₂ (C2/c)	stable	4.513	0.018
LiN ₃ *	mp-2659	C2/m	LiN ₃ $\rightarrow$ Li (R-3m) + 1.5 N ₂ (-)	stable	4.824	-
LiP ₅ *	mp-2412	Pna2_1	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.904	0.007
LiP ₅ *	mp-2412	Pna2_1	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.904	0.007
LiP ₅	mp-32760	Pna2_1	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.869	0.005
LiP ₅	mp-32760	Pna2_1	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.869	0.005
LiRh ₃ *	mp-1001787	Imm2	LiRh ₃ $\rightarrow$ Li (R-3m) + 3 Rh (Fm-3m)	stable	-1.915	0.012
LiS ₄	mp-1143546	P1	LiS ₄ $\rightarrow$ Li (R-3m) + 4 S (Fddd)	stable	-0.090	-0.017

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
LiTl ₃	mp-1185253	I4/mmm	LiTl ₃ → Li (R-3m) + 3 Tl (P6 ₋ 3/mmc)	stable	-1.442	0.000
Mg ₁₅ B	mp-1023515	P-6m2	Mg ₁₅ B → Mg (P6 ₋ 3/mmc) + 0.06667 B (R-3m)	not viable	-0.718	-0.010
Mg ₁₅ Ni	mp-1023622	P-6m2	Mg ₁₅ Ni → Mg (P6 ₋ 3/mmc) + 0.06667 Ni (Fm-3m)	not viable	-0.281	-0.014
Mg ₂ Ge*	mp-250	Fd-3m	Mg ₂ Ge → Mg (P6 ₋ 3/mmc) + 0.5 Ge (Fd-3m)	not viable	-1.345	-0.006
Mg ₂ Ni*	mp-2137	P6 ₋ 222	Mg ₂ Ni → Mg (P6 ₋ 3/mmc) + 0.5 Ni (Fm-3m)	stable	-1.391	-0.005
Mg ₃ Al	mp-978271	Pm-3m	Mg ₃ Al → Mg (P6 ₋ 3/mmc) + 0.3333 Al (Fm-3m)	not viable	-2.984	-0.009
Mg ₃ Ru ₂ *	mp-625	P4 ₋ 132	Mg ₃ Ru ₂ → Mg (P6 ₋ 3/mmc) + 0.6667 Ru (P6 ₋ 3/mmc)	stable	-1.928	-0.005
Mg ₃ Zr	mp-1094466	P6 ₋ 3/mmc	Mg ₃ Zr → Mg (P6 ₋ 3/mmc) + 0.3333 Zr (P6 ₋ 3/mmc)	not viable	-1.695	-0.004
Mg ₅ Sc	mp-1185837	R32	Mg ₅ Sc → Mg (P6 ₋ 3/mmc) + 0.2 Sc (P6 ₋ 3/mmc)	not viable	-0.471	-0.007
Mg ₅ Te	mp-1185828	R32	Mg ₅ Te → Mg (P6 ₋ 3/mmc) + 0.2 Te (P3 ₋ 121)	stable	-0.167	-0.014
MgAl ₂ *	mp-1102064	I4 ₋ 1/amd	MgAl ₂ → Mg (P6 ₋ 3/mmc) + 2 Al (Fm-3m)	not viable	-6.896	-0.000
MgCd*	mp-11307	Pmma	MgCd → Mg (P6 ₋ 3/mmc) + Cd (P6 ₋ 3/mmc)	stable	-2.137	0.000
MgCl ₂ *	mp-570259	P-3m1	MgCl ₂ → Mg (P6 ₋ 3/mmc) + Cl ₂ (Cmce)	stable	0.154	0.024
MgCl ₂ *	mp-23210	R-3m	MgCl ₂ → Mg (P6 ₋ 3/mmc) + Cl ₂ (Cmce)	stable	0.157	0.025
MgCl ₂	mp-569626	Ama2	MgCl ₂ → Mg (P6 ₋ 3/mmc) + Cl ₂ (Cmce)	stable	-0.086	0.034
MgF ₂ *	mp-1072956	Pnnm	MgF ₂ → Mg (P6 ₋ 3/mmc) + F ₂ (C2/c)	stable	0.001	0.021
MgF ₂	mp-1062842	Amm2	MgF ₂ → Mg (P6 ₋ 3/mmc) + F ₂ (C2/c)	stable	0.023	0.021
MgF ₂	mp-560236	C2/m	MgF ₂ → Mg (P6 ₋ 3/mmc) + F ₂ (C2/c)	stable	-0.391	0.022
MgH ₂ *	mp-23711	Pbcn	MgH ₂ → Mg (P6 ₋ 3/mmc) + H ₂ (P2 ₋ 12 ₋ 12 ₋ 1)	stable	-0.441	-0.001
MgIn ₂ *	mp-1077016	Fd-3m	MgIn ₂ → Mg (P6 ₋ 3/mmc) + 2 In (I4/mmm)	not viable	-4.702	-0.003
MgO*	mp-549706	P6 ₋ 3mc	MgO → Mg (P6 ₋ 3/mmc) + 0.5 O ₂ (C2/m)	stable	-1.418	0.011
MgO	mp-1009127	Pm-3m	MgO → Mg (P6 ₋ 3/mmc) + 0.5 O ₂ (C2/m)	stable	-1.076	0.005
MgP ₄ *	mp-384	P2 ₋ 1/c	MgP ₄ → Mg (P6 ₋ 3/mmc) + 4 P (P2/c)	stable	-0.107	0.001
MgPb ₂	mp-2016042	C2/m	MgPb ₂ → Mg (P6 ₋ 3/mmc) + 2 Pb (Fm-3m)	stable	-2.725	-0.007
MgPd ₃ *	mp-7753	Pm-3m	MgPd ₃ → Mg (P6 ₋ 3/mmc) + 3 Pd (Fm-3m)	stable	-5.600	0.026
MgS*	mp-1315	Fm-3m	MgS → Mg (P6 ₋ 3/mmc) + S (Fddd)	stable	-0.340	-0.001
MgS	mp-13032	F-43m	MgS → Mg (P6 ₋ 3/mmc) + S (Fddd)	stable	-0.416	0.000
MgSn	mp-1094669	Pm-3m	MgSn → Mg (P6 ₋ 3/mmc) + Sn (I4/mmm)	stable	-1.385	-0.013

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
MgSn	mp-1094801	P4/mmm	$\text{MgSn} \rightarrow \text{Mg (P6}_3/\text{mmc)} + \text{Sn (I4/mmm)}$	stable	-0.896	-0.012
MgTi	mp-1094346	P2_1/m	$\text{MgTi} \rightarrow \text{Mg (P6}_3/\text{mmc)} + \text{Ti (P6}_3/\text{mmc)}$	not viable	-1.599	0.001
MgTl ₃	mp-978262	P6_3/mmc	$\text{MgTl}_3 \rightarrow \text{Mg (P6}_3/\text{mmc)} + 3 \text{ Tl (P6}_3/\text{mmc)}$	not viable	-2.186	0.000
Mn ₂ O ₇ *	mp-28338	P2_1/c	$\text{Mn}_2\text{O}_7 \rightarrow \text{Mn (I-43m)} + 1.75 \text{ O}_2 \text{ (C2/m)}$	stable	-1.469	0.018
Mn ₄ B	mp-1221769	F222	$\text{Mn}_4\text{B} \rightarrow \text{Mn (I-43m)} + 0.25 \text{ B (R-3m)}$	stable	1.299	0.003
Mn ₄ Ga	mp-1221812	R-3m	$\text{Mn}_4\text{Ga} \rightarrow \text{Mn (I-43m)} + 0.25 \text{ Ga (Cmce)}$	meta-viable	5.276	0.003
MnAl ₁₂ *	mp-1104165	Im-3	$\text{MnAl}_{12} \rightarrow \text{Mn (I-43m)} + 12 \text{ Al (Fm-3m)}$	stable	-8.175	0.000
MnN	mp-1009130	F-43m	$\text{MnN} \rightarrow \text{Mn (I-43m)} + 0.5 \text{ N}_2 \text{ (-)}$	stable	2.122	-
MnO ₂ *	mp-510408	P4_2/mnm	$\text{MnO}_2 \rightarrow \text{Mn (I-43m)} + \text{O}_2 \text{ (C2/m)}$	stable	-0.527	0.015
Mo ₂ C	mp-1221498	P-3m1	$\text{Mo}_2\text{C} \rightarrow \text{Mo (Im-3m)} + 0.5 \text{ C (P6}_3/\text{mmc)}$	stable	-4.995	0.001
Mo ₃ Se ₄ *	mp-21021	R-3	$\text{Mo}_3\text{Se}_4 \rightarrow \text{Mo (Im-3m)} + 1.333 \text{ Se (P3}_3/\text{121)}$	stable	-2.810	0.008
MoPd ₂	mp-1206704	Immm	$\text{MoPd}_2 \rightarrow \text{Mo (Im-3m)} + 2 \text{ Pd (Fm-3m)}$	stable	-5.540	0.028
MoRu	mp-1221386	P-6m2	$\text{MoRu} \rightarrow \text{Mo (Im-3m)} + \text{Ru (P6}_3/\text{mmc)}$	not viable	-2.280	0.008
MoSe ₂ *	mp-1634	P6_3/mmc	$\text{MoSe}_2 \rightarrow \text{Mo (Im-3m)} + 2 \text{ Se (P3}_3/\text{121)}$	stable	-0.947	0.014
N ₂ O*	mp-1180473	Pnma	$\text{N}_2\text{O} \rightarrow \text{N}_2 \text{ (-)} + 0.5 \text{ O}_2 \text{ (C2/m)}$	stable	3.638	-
NO	mp-1180334	P2_12_12_1	$\text{NO} \rightarrow \text{N}_2 \text{ (-)} + \text{O}_2 \text{ (C2/m)}$	meta-viable	2.137	-
Na ₂ Cl	mp-1077015	Cmmm	$\text{Na}_2\text{Cl} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.25 \text{ Cl}_2 \text{ (Cmce)}$	stable	-0.397	0.025
Na ₂ Cl	mp-990084	R-3m	$\text{Na}_2\text{Cl} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.25 \text{ Cl}_2 \text{ (Cmce)}$	stable	-0.247	0.024
Na ₂ O*	mp-2352	Fm-3m	$\text{Na}_2\text{O} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.25 \text{ O}_2 \text{ (C2/m)}$	stable	-1.306	0.015
Na ₂ O	mp-1975297	C2/m	$\text{Na}_2\text{O} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.25 \text{ O}_2 \text{ (C2/m)}$	stable	-1.562	0.015
Na ₃ Cl	mp-1064484	P4/mmm	$\text{Na}_3\text{Cl} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.1667 \text{ Cl}_2 \text{ (Cmce)}$	stable	-0.138	0.020
Na ₃ Hg	mp-1186050	I4/mmm	$\text{Na}_3\text{Hg} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.3333 \text{ Hg (P6/mmm)}$	stable	-1.024	0.005
Na ₃ In	mp-864758	I4/mmm	$\text{Na}_3\text{In} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.3333 \text{ In (I4/mmm)}$	stable	-2.450	0.005
Na ₃ Mg	mp-982637	I4/mmm	$\text{Na}_3\text{Mg} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.3333 \text{ Mg (P6}_3/\text{mmc)}$	not viable	-0.077	-0.001
Na ₃ Si ₃₄	mp-1221235	R-3m	$\text{Na}_3\text{Si}_{34} \rightarrow \text{Na (P6}_3/\text{mmc)} + 11.33 \text{ Si (Fd-3m)}$	not viable	-0.427	-0.001
NaCl*	mp-22862	Fm-3m	$\text{NaCl} \rightarrow \text{Na (P6}_3/\text{mmc)} + 0.5 \text{ Cl}_2 \text{ (Cmce)}$	stable	0.374	0.032
NaN ₃ *	mp-1066400	R3m	$\text{NaN}_3 \rightarrow \text{Na (P6}_3/\text{mmc)} + 1.5 \text{ N}_2 \text{ (-)}$	stable	1.894	-
NaN ₃ *	mp-22003	R-3m	$\text{NaN}_3 \rightarrow \text{Na (P6}_3/\text{mmc)} + 1.5 \text{ N}_2 \text{ (-)}$	stable	2.173	-
NaN ₃ *	mp-1066400	R3m	$\text{NaN}_3 \rightarrow \text{Na (P6}_3/\text{mmc)} + 1.5 \text{ N}_2 \text{ (-)}$	stable	1.894	-

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
NaN_3^*	mp-22003	R-3m	$\text{NaN}_3 \rightarrow \text{Na (P6_3/mmc)} + 1.5 \text{ N}_2 (-)$	stable	2.173	—
NaN_3	mp-634410	C2	$\text{NaN}_3 \rightarrow \text{Na (P6_3/mmc)} + 1.5 \text{ N}_2 (-)$	meta-viable	2.902	—
NaN_3	mp-1065265	C2/m	$\text{NaN}_3 \rightarrow \text{Na (P6_3/mmc)} + 1.5 \text{ N}_2 (-)$	not viable	-1.732	—
NaP_{15}^*	mp-1194653	P-1	$\text{NaP}_{15} \rightarrow \text{Na (P6_3/mmc)} + 15 \text{ P (P2/c)}$	stable	-0.067	0.002
NaP_5^*	mp-31086	Pnma	$\text{NaP}_5 \rightarrow \text{Na (P6_3/mmc)} + 5 \text{ P (P2/c)}$	stable	-0.020	0.000
NaS^*	mp-2400	P6_3/mmc	$\text{NaS} \rightarrow \text{Na (P6_3/mmc)} + \text{S (Fddd)}$	stable	-0.495	0.005
NaS^*	mp-409	P-62m	$\text{NaS} \rightarrow \text{Na (P6_3/mmc)} + \text{S (Fddd)}$	stable	-0.491	0.003
NaS	mp-1180106	C2/m	$\text{NaS} \rightarrow \text{Na (P6_3/mmc)} + \text{S (Fddd)}$	not viable	-2.400	0.020
Nb_2O_5^*	mp-1595	C2/m	$\text{Nb}_2\text{O}_5 \rightarrow \text{Nb (Im-3m)} + 1.25 \text{ O}_2 \text{ (C2/m)}$	stable	-2.069	0.019
Nb_3Ir^*	mp-1458	Pm-3n	$\text{Nb}_3\text{Ir} \rightarrow \text{Nb (Im-3m)} + 0.3333 \text{ Ir (Fm-3m)}$	stable	-10.122	0.005
Nb_3Ni	mp-999394	I4/mmm	$\text{Nb}_3\text{Ni} \rightarrow \text{Nb (Im-3m)} + 0.3333 \text{ Ni (Fm-3m)}$	not viable	-6.850	0.008
Nb_4Rh	mp-1220441	Fmmm	$\text{Nb}_4\text{Rh} \rightarrow \text{Nb (Im-3m)} + 0.25 \text{ Rh (Fm-3m)}$	stable	-4.876	0.007
NbCd_3^*	mp-11309	Pm-3m	$\text{NbCd}_3 \rightarrow \text{Nb (Im-3m)} + 3 \text{ Cd (P6_3/mmc)}$	not viable	-6.465	0.000
NbGa_3^*	mp-1973	I4/mmm	$\text{NbGa}_3 \rightarrow \text{Nb (Im-3m)} + 3 \text{ Ga (Cmce)}$	stable	-0.757	0.000
NbO^*	mp-2692	Fm-3m	$\text{NbO} \rightarrow \text{Nb (Im-3m)} + 0.5 \text{ O}_2 \text{ (C2/m)}$	stable	-2.318	0.010
NbO	mp-634965	P4/mmm	$\text{NbO} \rightarrow \text{Nb (Im-3m)} + 0.5 \text{ O}_2 \text{ (C2/m)}$	stable	-3.149	0.015
NbP^*	mp-9830	I4_1/amd	$\text{NbP} \rightarrow \text{Nb (Im-3m)} + \text{P (P2/c)}$	stable	-2.472	0.004
NbP	mp-1220317	Cmmm	$\text{NbP} \rightarrow \text{Nb (Im-3m)} + \text{P (P2/c)}$	stable	-1.173	0.002
NbRh_2	mp-1220414	Pm	$\text{NbRh}_2 \rightarrow \text{Nb (Im-3m)} + 2 \text{ Rh (Fm-3m)}$	stable	-5.273	0.014
NbS_2	mp-774873	P6_3/mmc	$\text{NbS}_2 \rightarrow \text{Nb (Im-3m)} + 2 \text{ S (Fddd)}$	stable	-1.997	0.019
NbZn_{16}^*	mp-640041	Cmcm	$\text{NbZn}_{16} \rightarrow \text{Nb (Im-3m)} + 16 \text{ Zn (P6_3/mmc)}$	stable	-7.122	0.000
Ni_3Mo^*	mp-11506	Pmmn	$\text{Ni}_3\text{Mo} \rightarrow \text{Ni (Fm-3m)} + 0.3333 \text{ Mo (Im-3m)}$	stable	-4.295	0.007
NiB^*	mp-14019	Cmcm	$\text{NiB} \rightarrow \text{Ni (Fm-3m)} + \text{B (R-3m)}$	stable	-0.809	0.013
NiHg_4^*	mp-570835	Im-3m	$\text{NiHg}_4 \rightarrow \text{Ni (Fm-3m)} + 4 \text{ Hg (P6/mmm)}$	not viable	-0.361	0.006
NiS^*	mp-1547	R3m	$\text{NiS} \rightarrow \text{Ni (Fm-3m)} + \text{S (Fddd)}$	stable	-1.148	0.017
NiSb_3^*	mp-1106023	Im-3	$\text{NiSb}_3 \rightarrow \text{Ni (Fm-3m)} + 3 \text{ Sb (R-3m)}$	stable	-0.214	0.002
OsC^*	mp-7142	P-6m2	$\text{OsC} \rightarrow \text{Os (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.017	-0.001
OsC	mp-1009537	Pm-3m	$\text{OsC} \rightarrow \text{Os (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.318	-0.002
OsC	mp-1009539	Fm-3m	$\text{OsC} \rightarrow \text{Os (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-3.114	-0.009
OsC	mp-999308	P6_3/mmc	$\text{OsC} \rightarrow \text{Os (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.957	-0.005

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
P_4Se_5^*	mp-2447	Pna2_1	$\text{P}_4\text{Se}_5 \rightarrow \text{P (P2/c)} + 1.25 \text{ Se (P3_121)}$	stable	0.356	-0.000
PI_2^*	mp-29443	P-1	$\text{PI}_2 \rightarrow \text{P (P2/c)} + 2 \text{ I (Cmce)}$	stable	-0.086	0.015
PNO	mp-675742	P-1	$\text{PNO} \rightarrow \text{P (P2/c)} + 0.5 \text{ N}_2 (-) + 0.5 \text{ O}_2 (\text{C2/m})$	meta-viable	0.674	—
PS^*	mp-612	C2/c	$\text{PS} \rightarrow \text{P (P2/c)} + \text{S (Fddd)}$	stable	-0.025	0.008
PW_3	mp-1209828	I-42m	$\text{PW}_3 \rightarrow \text{P (P2/c)} + 3 \text{ W (Im-3m)}$	stable	-11.879	0.002
Pb_2Au^*	mp-22795	I4/mcm	$\text{Pb}_2\text{Au} \rightarrow \text{Pb (Fm-3m)} + 0.5 \text{ Au (Fm-3m)}$	stable	-6.812	0.010
PbAu_2^*	mp-19871	Fd-3m	$\text{PbAu}_2 \rightarrow \text{Pb (Fm-3m)} + 2 \text{ Au (Fm-3m)}$	stable	-10.943	0.005
PbAu_3	mp-982246	Fm-3m	$\text{PbAu}_3 \rightarrow \text{Pb (Fm-3m)} + 3 \text{ Au (Fm-3m)}$	not viable	-10.804	0.000
PbN_6^*	mp-620058	Pna2_1	$\text{PbN}_6 \rightarrow \text{Pb (Fm-3m)} + 3 \text{ N}_2 (-)$	meta-viable	1.685	—
PbN_6^*	mp-667338	Pnma	$\text{PbN}_6 \rightarrow \text{Pb (Fm-3m)} + 3 \text{ N}_2 (-)$	meta-viable	1.695	—
PbSe^*	mp-22009	Fmm2	$\text{PbSe} \rightarrow \text{Pb (Fm-3m)} + \text{Se (P3_121)}$	stable	-2.952	-0.006
PbSe^*	mp-2201	Fm-3m	$\text{PbSe} \rightarrow \text{Pb (Fm-3m)} + \text{Se (P3_121)}$	stable	-2.482	-0.003
PtCl_2	mp-1219748	C2/m	$\text{PtCl}_2 \rightarrow \text{Pt (Fm-3m)} + \text{Cl}_2 (\text{Cmce})$	stable	-1.841	0.023
Rb_2Hg_7^*	mp-31474	P-3m1	$\text{Rb}_2\text{Hg}_7 \rightarrow \text{Rb (Im-3m)} + 3.5 \text{ Hg (P6/mmm)}$	stable	-1.184	0.002
Rb_2O^*	mp-1394	Fm-3m	$\text{Rb}_2\text{O} \rightarrow \text{Rb (Im-3m)} + 0.25 \text{ O}_2 (\text{C2/m})$	stable	0.724	0.010
Rb_2O	mp-2023482	P-3m1	$\text{Rb}_2\text{O} \rightarrow \text{Rb (Im-3m)} + 0.25 \text{ O}_2 (\text{C2/m})$	stable	-0.759	0.005
Rb_3In	mp-974948	Pm-3m	$\text{Rb}_3\text{In} \rightarrow \text{Rb (Im-3m)} + 0.3333 \text{ In (I4/mmm)}$	not viable	-2.599	0.000
Rb_3Li	mp-974938	Pm-3m	$\text{Rb}_3\text{Li} \rightarrow \text{Rb (Im-3m)} + 0.3333 \text{ Li (R-3m)}$	not viable	-0.756	0.003
Rb_3P	mp-974822	Fm-3m	$\text{Rb}_3\text{P} \rightarrow \text{Rb (Im-3m)} + 0.3333 \text{ P (P2/c)}$	stable	0.907	-0.002
Rb_3P_{11}	mp-1173444	C2/m	$\text{Rb}_3\text{P}_{11} \rightarrow \text{Rb (Im-3m)} + 3.667 \text{ P (P2/c)}$	stable	-0.000	-0.001
RbCl^*	mp-23299	Pm-3m	$\text{RbCl} \rightarrow \text{Rb (Im-3m)} + 0.5 \text{ Cl}_2 (\text{Cmce})$	stable	1.792	0.027
RbF^*	mp-2064	Pm-3m	$\text{RbF} \rightarrow \text{Rb (Im-3m)} + 0.5 \text{ F}_2 (\text{C2/c})$	stable	3.924	0.019
RbF	mp-11718	Fm-3m	$\text{RbF} \rightarrow \text{Rb (Im-3m)} + 0.5 \text{ F}_2 (\text{C2/c})$	stable	1.950	0.019
RbI^*	mp-22903	Fm-3m	$\text{RbI} \rightarrow \text{Rb (Im-3m)} + \text{I (Cmce)}$	stable	0.819	0.033
RbN_3^*	mp-581833	P4/mmm	$\text{RbN}_3 \rightarrow \text{Rb (Im-3m)} + 1.5 \text{ N}_2 (-)$	stable	2.345	—
RbN_3	mp-1219567	Amm2	$\text{RbN}_3 \rightarrow \text{Rb (Im-3m)} + 1.5 \text{ N}_2 (-)$	meta-viable	0.129	—
RbP^*	mp-1179688	C2/m	$\text{RbP} \rightarrow \text{Rb (Im-3m)} + \text{P (P2/c)}$	not viable	-1.931	0.007
RbP	mp-1058499	Immm	$\text{RbP} \rightarrow \text{Rb (Im-3m)} + \text{P (P2/c)}$	not viable	-1.910	0.008
RbP_{15}	mp-1209298	P-1	$\text{RbP}_{15} \rightarrow \text{Rb (Im-3m)} + 15 \text{ P (P2/c)}$	stable	-0.107	0.002
RbS^*	mp-558071	Immm	$\text{RbS} \rightarrow \text{Rb (Im-3m)} + \text{S (Fddd)}$	stable	-0.322	-0.041
RbS	mp-1179684	C2/m	$\text{RbS} \rightarrow \text{Rb (Im-3m)} + \text{S (Fddd)}$	stable	-2.735	-0.029
Re_2Si^*	mp-669547	P2_1/c	$\text{Re}_2\text{Si} \rightarrow \text{Re (P6_3/mmc)} + 0.5 \text{ Si (Fd-3m)}$	not viable	-2.647	-0.001

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
Re_3Ge_7^*	mp-29141	Cmcm	$\text{Re}_3\text{Ge}_7 \rightarrow \text{Re (P6_3/mmc)} + 2.333 \text{ Ge (Fd-3m)}$	not viable	-0.246	0.002
Re_3N_5	mp-1080194	Imm2	$\text{Re}_3\text{N}_5 \rightarrow \text{Re (P6_3/mmc)} + 0.8333 \text{ N}_2 (-)$	not viable	-2.055	—
Re_3Pd	mp-862586	P6_3/mmc	$\text{Re}_3\text{Pd} \rightarrow \text{Re (P6_3/mmc)} + 0.3333 \text{ Pd (Fm-3m)}$	stable	-3.849	0.015
Re_5Si_3^*	mp-1105959	I4/mcm	$\text{Re}_5\text{Si}_3 \rightarrow \text{Re (P6_3/mmc)} + 0.6 \text{ Si (Fd-3m)}$	stable	-1.914	-0.004
ReC^*	mp-1079494	P6_3/mmc	$\text{ReC} \rightarrow \text{Re (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.662	-0.002
ReC	mp-1009731	Pm-3m	$\text{ReC} \rightarrow \text{Re (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-1.996	-0.000
RuC^*	mp-10030	P-6m2	$\text{RuC} \rightarrow \text{Ru (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.130	0.002
RuC	mp-1009787	Fm-3m	$\text{RuC} \rightarrow \text{Ru (P6_3/mmc)} + \text{C (P6_3/mmc)}$	not viable	-2.860	-0.006
RuRh_3	mp-974335	Pm-3m	$\text{RuRh}_3 \rightarrow \text{Ru (P6_3/mmc)} + 3 \text{ Rh (Fm-3m)}$	not viable	-4.136	0.008
S_2N	—	P4_2nm	$\text{S}_2\text{N} \rightarrow \text{S (Fddd)} + 0.25 \text{ N}_2 (-)$	n/a	0.665	—
SN	—	P2_1/c	$\text{SN} \rightarrow \text{S (Fddd)} + 0.5 \text{ N}_2 (-)$	n/a	1.066	—
SN^*	mp-236	P2_1/c	$\text{SN} \rightarrow \text{S (Fddd)} + 0.5 \text{ N}_2 (-)$	meta-viable	0.144	—
SN	mp-1078816	P2_1/c	$\text{SN} \rightarrow \text{S (Fddd)} + 0.5 \text{ N}_2 (-)$	not viable	-1.246	—
SN	mp-1173453	P2_1	$\text{SN} \rightarrow \text{S (Fddd)} + 0.5 \text{ N}_2 (-)$	not viable	-0.705	—
SN	mp-684642	P2_1	$\text{SN} \rightarrow \text{S (Fddd)} + 0.5 \text{ N}_2 (-)$	meta-viable	0.726	—
Sb_2S_5	mp-1359089	P2_1/c	$\text{Sb}_2\text{S}_5 \rightarrow \text{Sb (R-3m)} + 2.5 \text{ S (Fddd)}$	stable	-0.457	-0.004
Sc_2O_3	mp-755313	R-3c	$\text{Sc}_2\text{O}_3 \rightarrow \text{Sc (P6_3/mmc)} + 0.75 \text{ O}_2 (\text{C2/m})$	stable	0.083	0.018
Sc_5Ge_3^*	mp-17190	P6_3/mcm	$\text{Sc}_5\text{Ge}_3 \rightarrow \text{Sc (P6_3/mmc)} + 0.6 \text{ Ge (Fd-3m)}$	stable	-0.525	0.000
ScAg_2^*	mp-1018128	I4/mmm	$\text{ScAg}_2 \rightarrow \text{Sc (P6_3/mmc)} + 2 \text{ Ag (Fm-3m)}$	stable	-2.684	0.000
ScOs_3	mp-981384	Pm-3m	$\text{ScOs}_3 \rightarrow \text{Sc (P6_3/mmc)} + 3 \text{ Os (P6_3/mmc)}$	not viable	-2.673	0.006
ScTc_3	mp-867262	Fm-3m	$\text{ScTc}_3 \rightarrow \text{Sc (P6_3/mmc)} + 3 \text{ Tc (P6_3/mmc)}$	stable	-2.445	0.003
ScTe^*	mp-10026	P6_3/mmc	$\text{ScTe} \rightarrow \text{Sc (P6_3/mmc)} + \text{Te (P3_121)}$	stable	0.902	0.013
Si_3H	mp-995193	P-3m1	$\text{Si}_3\text{H} \rightarrow \text{Si (Fd-3m)} + 0.1667 \text{ H}_2 (\text{P2_12_12_1})$	not viable	-0.040	-0.000
Si_3Pt	mp-972763	P6_3/mmc	$\text{Si}_3\text{Pt} \rightarrow \text{Si (Fd-3m)} + 0.3333 \text{ Pt (Fm-3m)}$	stable	-0.484	0.009
Si_3W_5^*	mp-31219	I4/mcm	$\text{Si}_3\text{W}_5 \rightarrow \text{Si (Fd-3m)} + 1.667 \text{ W (Im-3m)}$	stable	-10.167	0.000
SiAu_3	mp-1186998	Fm-3m	$\text{SiAu}_3 \rightarrow \text{Si (Fd-3m)} + 3 \text{ Au (Fm-3m)}$	not viable	-10.873	-0.005
SiH_2^*	mp-1192532	P2_1/m	$\text{SiH}_2 \rightarrow \text{Si (Fd-3m)} + \text{H}_2 (\text{P2_12_12_1})$	not viable	-0.098	0.000
SiO_2^*	mp-546794	I-42d	$\text{SiO}_2 \rightarrow \text{Si (Fd-3m)} + \text{O}_2 (\text{C2/m})$	stable	2.142	0.018
SiO_2^*	mp-9258	Pa-3	$\text{SiO}_2 \rightarrow \text{Si (Fd-3m)} + \text{O}_2 (\text{C2/m})$	stable	3.333	0.014
SiO_2	mp-10064	Fm-3m	$\text{SiO}_2 \rightarrow \text{Si (Fd-3m)} + \text{O}_2 (\text{C2/m})$	stable	2.173	0.017
SiO_2	mp-556044	P-31c	$\text{SiO}_2 \rightarrow \text{Si (Fd-3m)} + \text{O}_2 (\text{C2/m})$	stable	0.096	0.016
SiO_2	mp-638049	F4_132	$\text{SiO}_2 \rightarrow \text{Si (Fd-3m)} + \text{O}_2 (\text{C2/m})$	stable	0.385	0.009
SiRh^*	mp-2287	P2_1/c	$\text{SiRh} \rightarrow \text{Si (Fd-3m)} + \text{Rh (Fm-3m)}$	stable	-1.038	0.012
SiS_2^*	mp-1095294	P2_1/c	$\text{SiS}_2 \rightarrow \text{Si (Fd-3m)} + 2 \text{ S (Fddd)}$	stable	0.996	0.019

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
SiTc ₃	mp-972784	P6 ₃ /mmc	SiTc ₃ → Si (Fd-3m) + 3 Tc (P6 ₃ /mmc)	stable	-2.525	-0.002
SiTe ₂ *	mp-2755	P-3m1	SiTe ₂ → Si (Fd-3m) + 2 Te (P3 ₁₂₁)	meta-viable	0.560	-0.009
Sn ₃ S	mp-1187038	Fm-3m	Sn ₃ S → Sn (I4/mmm) + 0.3333 S (Fddd)	not viable	-4.045	0.008
Sn ₃ Se	mp-1187055	Fm-3m	Sn ₃ Se → Sn (I4/mmm) + 0.3333 Se (P3 ₁₂₁)	not viable	-3.890	0.005
Sn ₇ Ir ₅ *	mp-20466	I4mm	Sn ₇ Ir ₅ → Sn (I4/mmm) + 0.7143 Ir (Fm-3m)	stable	-6.974	0.012
Sn ₇ Pd ₅	mp-1219006	C2	Sn ₇ Pd ₅ → Sn (I4/mmm) + 0.7143 Pd (Fm-3m)	stable	-2.751	0.021
SnBr ₂	mp-978985	P4 ₂ /mnm	SnBr ₂ → Sn (I4/mmm) + 2 Br (Cmce)	stable	-1.529	0.007
SnP*	mp-475	Fm-3m	SnP → Sn (I4/mmm) + P (P2/c)	not viable	-1.202	0.005
SnPb ₃	mp-1187118	P6 ₃ /mmc	SnPb ₃ → Sn (I4/mmm) + 3 Pb (Fm-3m)	not viable	-5.196	0.006
SnPt*	mp-19856	P6 ₃ /mmc	SnPt → Sn (I4/mmm) + Pt (Fm-3m)	stable	-5.180	0.025
Sr ₂ As*	mp-15697	I4/mmm	Sr ₂ As → Sr (P6 ₃ /mmc) + 0.5 As (Cmce)	stable	-0.667	0.037
Sr ₃ As ₄	mp-678007	P1	Sr ₃ As ₄ → Sr (P6 ₃ /mmc) + 1.333 As (Cmce)	stable	-0.746	0.013
Sr ₃ N ₂	mp-1013529	Pm-3m	Sr ₃ N ₂ → Sr (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	-2.433	—
Sr ₃ P ₄ *	mp-14288	Fdd2	Sr ₃ P ₄ → Sr (P6 ₃ /mmc) + 1.333 P (P2/c)	stable	-1.047	0.019
Sr ₃ P ₄	mp-682953	P1	Sr ₃ P ₄ → Sr (P6 ₃ /mmc) + 1.333 P (P2/c)	stable	-1.079	0.019
Sr ₃ Sn	mp-1187136	Pm-3m	Sr ₃ Sn → Sr (P6 ₃ /mmc) + 0.3333 Sn (I4/mmm)	stable	-1.601	0.023
Sr ₅ Sn ₃ *	mp-17720	I4/mcm	Sr ₅ Sn ₃ → Sr (P6 ₃ /mmc) + 0.6 Sn (I4/mmm)	stable	-2.198	0.021
SrAl ₂ *	mp-1638	Fd-3m	SrAl ₂ → Sr (P6 ₃ /mmc) + 2 Al (Fm-3m)	stable	-7.322	0.028
SrB ₂	mp-1008755	P6/mmm	SrB ₂ → Sr (P6 ₃ /mmc) + 2 B (R-3m)	not viable	-2.789	0.028
SrF ₂ *	mp-1102240	Pnma	SrF ₂ → Sr (P6 ₃ /mmc) + F ₂ (C2/c)	stable	-0.542	0.040
SrF ₂	mp-1102911	Pnma	SrF ₂ → Sr (P6 ₃ /mmc) + F ₂ (C2/c)	stable	-0.148	0.041
SrN*	mp-1058586	Fm-3m	SrN → Sr (P6 ₃ /mmc) + 0.5 N ₂ (-)	stable	-2.078	—
SrN ₆ *	mp-2131	Fddd	SrN ₆ → Sr (P6 ₃ /mmc) + 3 N ₂ (-)	stable	1.872	—
SrO ₁₀	mp-1208655	P4/mmm	SrO ₁₀ → Sr (P6 ₃ /mmc) + 5 O ₂ (C2/m)	not viable	-2.680	0.018
SrO ₂ *	mp-2697	I4/mmm	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.854	0.026
SrO ₂ *	mp-726705	Pnma	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-1.921	0.014
SrO ₂	mp-2763179	Pnma	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-1.752	0.015
SrS*	mp-10627	Pm-3m	SrS → Sr (P6 ₃ /mmc) + S (Fddd)	stable	-0.109	0.026

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHP}_{\text{ab.}}$
SrSi*	mp-2661	Cmcm	SrSi $\rightarrow$ Sr (P6 ₃ /mmc) + Si (Fd-3m)	stable	-1.341	0.031
SrTl ₃	mp-1206636	Pm-3m	SrTl ₃ $\rightarrow$ Sr (P6 ₃ /mmc) + 3 Tl (P6 ₃ /mmc)	stable	-2.206	0.013
Ta ₄ C ₃	mp-1218000	R3m	Ta ₄ C ₃ $\rightarrow$ Ta (Im-3m) + 0.75 C (P6 ₃ /mmc)	stable	-5.556	0.000
Ta ₅ Ge ₃ *	mp-17593	I4/mcm	Ta ₅ Ge ₃ $\rightarrow$ Ta (Im-3m) + 0.6 Ge (Fd-3m)	stable	-5.087	0.000
Ta ₇ Co ₆ *	mp-1104548	R-3m	Ta ₇ Co ₆ $\rightarrow$ Ta (Im-3m) + 0.8571 Co (Fm-3m)	stable	-9.406	0.002
TaBe ₁₂ *	mp-567842	I4/mmm	TaBe ₁₂ $\rightarrow$ Ta (Im-3m) + 12 Be (P6 ₃ /mmc)	stable	-2.228	0.000
TaN*	mp-1497	P6/mmm	TaN $\rightarrow$ Ta (Im-3m) + 0.5 N ₂ (-)	stable	-3.469	-
TaN	mp-1009831	Pm-3m	TaN $\rightarrow$ Ta (Im-3m) + 0.5 N ₂ (-)	stable	-1.592	-
TaN	mp-1009833	F-43m	TaN $\rightarrow$ Ta (Im-3m) + 0.5 N ₂ (-)	stable	-2.877	-
TaP*	mp-1067587	I4 ₁ md	TaP $\rightarrow$ Ta (Im-3m) + P (P2/c)	stable	-2.273	0.005
TaP	mp-1187244	P-6m2	TaP $\rightarrow$ Ta (Im-3m) + P (P2/c)	stable	-1.204	0.005
TaRe	mp-1217894	Cmmm	TaRe $\rightarrow$ Ta (Im-3m) + Re (P6 ₃ /mmc)	stable	-2.608	0.003
TaSb ₂ *	mp-11697	C2/m	TaSb ₂ $\rightarrow$ Ta (Im-3m) + 2 Sb (R-3m)	stable	-1.785	0.008
Tc ₃ Mo	mp-1187395	Pm-3m	Tc ₃ Mo $\rightarrow$ Tc (P6 ₃ /mmc) + 0.3333 Mo (Im-3m)	not viable	-4.944	0.004
Tc ₃ Os	mp-862779	P6 ₃ /mmc	Tc ₃ Os $\rightarrow$ Tc (P6 ₃ /mmc) + 0.3333 Os (P6 ₃ /mmc)	stable	-3.799	0.002
Tc ₃ W	mp-1187398	P6 ₃ /mmc	Tc ₃ W $\rightarrow$ Tc (P6 ₃ /mmc) + 0.3333 W (Im-3m)	not viable	-7.079	0.005
TcW	mp-1217337	Cmmm	TcW $\rightarrow$ Tc (P6 ₃ /mmc) + W (Im-3m)	not viable	-3.712	0.003
Te ₂ Au*	mp-567525	C2/m	Te ₂ Au $\rightarrow$ Te (P3 ₁₂₁) + 0.5 Au (Fm-3m)	stable	-2.231	-0.006
Te ₂ Ir*	mp-1551	C2/m	Te ₂ Ir $\rightarrow$ Te (P3 ₁₂₁) + 0.5 Ir (Fm-3m)	stable	-2.030	0.009
Te ₂ Pd*	mp-782	P-3m1	Te ₂ Pd $\rightarrow$ Te (P3 ₁₂₁) + 0.5 Pd (Fm-3m)	stable	1.378	0.015
Te ₂ Pd ₇	mp-1217443	R-3m	Te ₂ Pd ₇ $\rightarrow$ Te (P3 ₁₂₁) + 3.5 Pd (Fm-3m)	stable	-4.945	0.024
Te ₂ Ru*	mp-267	Pnnm	Te ₂ Ru $\rightarrow$ Te (P3 ₁₂₁) + 0.5 Ru (P6 ₃ /mmc)	stable	0.657	0.008
TeO ₂ *	mp-8377	P2 _{12_12_1}	TeO ₂ $\rightarrow$ Te (P3 ₁₂₁) + O ₂ (C2/m)	stable	-0.149	0.011
Ti ₂ Cu*	mp-742	I4/mmm	Ti ₂ Cu $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 Cu (Fm-3m)	stable	-0.718	0.002
Ti ₃ Ge*	mp-1189044	I-4	Ti ₃ Ge $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 Ge (Fd-3m)	stable	-1.362	0.000
Ti ₃ In*	mp-866046	P6 ₃ /mmc	Ti ₃ In $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 In (I4/mmm)	stable	-2.597	0.000
Ti ₃ Nb	mp-1217091	Amm2	Ti ₃ Nb $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 Nb (Im-3m)	not viable	-1.419	0.000
Ti ₃ Pt*	mp-2339	Pm-3n	Ti ₃ Pt $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 Pt (Fm-3m)	stable	-3.640	0.011
TiBe ₁₂ *	mp-1104067	I4/mmm	TiBe ₁₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 12 Be (P6 ₃ /mmc)	stable	-1.742	0.000
TiMo ₂	mp-1216675	Fmmm	TiMo ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Mo (Im-3m)	stable	-3.376	0.003

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
TiMo ₂	mp-1216675	Fmmm	TiMo ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Mo (Im-3m)	stable	-3.376	0.003
TiNi*	mp-1048	P2 ₁ /m	TiNi $\rightarrow$ Ti (P6 ₃ /mmc) + Ni (Fm-3m)	stable	-1.589	0.015
TiNi	mp-1067248	Cmcm	TiNi $\rightarrow$ Ti (P6 ₃ /mmc) + Ni (Fm-3m)	stable	-1.659	0.015
TiO*	mp-1071163	P-62m	TiO $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 O ₂ (C2/m)	stable	-0.141	0.015
TiO ₂ *	mp-1439	Pbcn	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.646	0.019
TiO ₂ *	mp-9173	Pnma	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.651	0.018
TiO ₂ *	mp-430	P2 ₁ /c	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.627	0.018
TiSi ₂ *	mp-1077503	Cmcm	TiSi ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Si (Fd-3m)	stable	0.880	0.000
TiSi ₂	mp-570991	Immm	TiSi ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Si (Fd-3m)	stable	1.334	0.000
TiW	mp-1216621	Cmmm	TiW $\rightarrow$ Ti (P6 ₃ /mmc) + W (Im-3m)	not viable	-3.460	0.000
Tl ₃ Au	mp-1080569	Pm-3n	Tl ₃ Au $\rightarrow$ Tl (P6 ₃ /mmc) + 0.3333 Au (Fm-3m)	not viable	-6.137	0.000
Tl ₃ Ga	mp-1187539	Fm-3m	Tl ₃ Ga $\rightarrow$ Tl (P6 ₃ /mmc) + 0.3333 Ga (Cmce)	not viable	-3.377	0.000
Tl ₄ Sn	mp-1216560	R-3m	Tl ₄ Sn $\rightarrow$ Tl (P6 ₃ /mmc) + 0.25 Sn (I4/mmm)	meta-viable	1.836	0.001
TlAs	mp-1007770	F-43m	TlAs $\rightarrow$ Tl (P6 ₃ /mmc) + As (Cmce)	not viable	-0.514	0.001
TlAu ₃	mp-1187508	Pm-3m	TlAu ₃ $\rightarrow$ Tl (P6 ₃ /mmc) + 3 Au (Fm-3m)	not viable	-11.142	0.000
TiC*	mp-567465	Fm-3m	TiC $\rightarrow$ Ti (P6 ₃ /mmc) + C (P6 ₃ /mmc)	not viable	-3.140	0.000
TiC	mp-1058887	P4/mmm	TiC $\rightarrow$ Ti (P6 ₃ /mmc) + C (P6 ₃ /mmc)	not viable	-3.364	0.000
TiCl*	mp-571079	Cmcm	TiCl $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 Cl ₂ (Cmce)	stable	-2.297	0.007
TiN*	mp-1017982	P6 ₃ /mmc	TiN $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 N ₂ (-)	not viable	-1.288	—
TiN	mp-1002222	Fm-3m	TiN $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 N ₂ (-)	meta-viable	0.139	—
TiN	mp-1007778	F-43m	TiN $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 N ₂ (-)	not viable	-0.733	—
VC ₃	mp-1095057	P6 ₃ /mmc	VC ₃ $\rightarrow$ V (Im-3m) + 3 Co (Fm-3m)	not viable	-8.838	0.004
VN	mp-1017532	P6 ₃ /mmc	VN $\rightarrow$ V (Im-3m) + 0.5 N ₂ (-)	stable	-1.629	—
VNi ₂ *	mp-11531	Immm	VNi ₂ $\rightarrow$ V (Im-3m) + 2 Ni (Fm-3m)	stable	-3.105	0.011
W ₉ C ₄	mp-684989	R32	W ₉ C ₄ $\rightarrow$ W (Im-3m) + 0.4444 C (P6 ₃ /mmc)	stable	-13.727	0.000
WBr ₅	mp-1436789	P-1	WBr ₅ $\rightarrow$ W (Im-3m) + 5 Br (Cmce)	not viable	-1.823	0.007
WC*	mp-13136	Fm-3m	WC $\rightarrow$ W (Im-3m) + C (P6 ₃ /mmc)	not viable	-6.268	-0.000
WC	mp-1008630	Pm-3m	WC $\rightarrow$ W (Im-3m) + C (P6 ₃ /mmc)	not viable	-5.940	0.000

Compound	mp-id	SG	Reaction (products, SG in parentheses)	Viab.	ΔICOHP	$\Delta\text{ICOHPab.}$
WC	mp-1008635	F-43m	$\text{WC} \rightarrow \text{W (Im-3m)} + \text{C (P6}_3/\text{mmc)}$	not viable	-7.732	0.000
WO_2^*	mp-1524394	P2_1/c	$\text{WO}_2 \rightarrow \text{W (Im-3m)} + \text{O}_2 \text{ (C2/m)}$	stable	-4.597	0.019
Y_2O_3^*	mp-558573	C2/m	$\text{Y}_2\text{O}_3 \rightarrow \text{Y (P6}_3/\text{mmc)} + 0.75 \text{O}_2 \text{ (C2/m)}$	stable	1.143	0.018
Y_3Co^*	mp-1105598	Pnma	$\text{Y}_3\text{Co} \rightarrow \text{Y (P6}_3/\text{mmc)} + 0.3333 \text{Co (Fm-3m)}$	stable	-2.910	0.001
Y_3Co	mp-1189329	P2_1	$\text{Y}_3\text{Co} \rightarrow \text{Y (P6}_3/\text{mmc)} + 0.3333 \text{Co (Fm-3m)}$	stable	-2.883	0.001
YAg_3	mp-1187811	Fm-3m	$\text{YAg}_3 \rightarrow \text{Y (P6}_3/\text{mmc)} + 3 \text{Ag (Fm-3m)}$	stable	-5.208	0.000
$\text{Zn}_{13}\text{Rh}^*$	mp-13447	C2/m	$\text{Zn}_{13}\text{Rh} \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.07692 \text{Rh (Fm-3m)}$	stable	-6.821	0.003
Zn_2Ag	mp-1093999	P6_3/mmc	$\text{Zn}_2\text{Ag} \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.5 \text{Ag (Fm-3m)}$	stable	-7.151	0.000
Zn_3Au	mp-1188000	P6_3/mmc	$\text{Zn}_3\text{Au} \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.3333 \text{Au (Fm-3m)}$	stable	-8.665	0.000
Zn_3N_2	mp-1524551	P6_3/mmc	$\text{Zn}_3\text{N}_2 \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.3333 \text{N}_2 \text{ (-)}$	stable	-4.296	—
Zn_3N_2^*	mp-9460	Ia-3	$\text{Zn}_3\text{N}_2 \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.3333 \text{N}_2 \text{ (-)}$	stable	-4.428	—
Zn_7Tc	mp-1206378	Fm-3m	$\text{Zn}_7\text{Tc} \rightarrow \text{Zn (P6}_3/\text{mmc)} + 0.1429 \text{Tc (P6}_3/\text{mmc)}$	stable	-6.233	0.001
ZnAg^*	mp-1912	Pm-3m	$\text{ZnAg} \rightarrow \text{Zn (P6}_3/\text{mmc)} + \text{Ag (Fm-3m)}$	stable	-4.578	0.000
ZnPb_3	mp-1187949	Fm-3m	$\text{ZnPb}_3 \rightarrow \text{Zn (P6}_3/\text{mmc)} + 3 \text{Pb (Fm-3m)}$	not viable	-5.826	0.012
ZnSn	—	P6_3/mmc	$\text{ZnSn} \rightarrow \text{Zn (P6}_3/\text{mmc)} + \text{Sn (I4/mmm)}$	n/a	-5.014	0.004
Zr_2Au^*	mp-2348	I4/mmm	$\text{Zr}_2\text{Au} \rightarrow \text{Zr (P6}_3/\text{mmc)} + 0.5 \text{Au (Fm-3m)}$	stable	-2.179	0.000
Zr_3Be	mp-1188016	I4/mmm	$\text{Zr}_3\text{Be} \rightarrow \text{Zr (P6}_3/\text{mmc)} + 0.3333 \text{Be (P6}_3/\text{mmc)}$	not viable	-1.880	0.000
Zr_3Fe^*	mp-31205	Cmcm	$\text{Zr}_3\text{Fe} \rightarrow \text{Zr (P6}_3/\text{mmc)} + 0.3333 \text{Fe (Im-3m)}$	stable	-1.584	0.003
Zr_3Ge	mp-1188039	P6_3/mmc	$\text{Zr}_3\text{Ge} \rightarrow \text{Zr (P6}_3/\text{mmc)} + 0.3333 \text{Ge (Fd-3m)}$	stable	-1.667	0.000
Zr_5Pb_3^*	mp-681992	P6_3/mcm	$\text{Zr}_5\text{Pb}_3 \rightarrow \text{Zr (P6}_3/\text{mmc)} + 0.6 \text{Pb (Fm-3m)}$	stable	-2.335	0.006
ZrBr^*	mp-1065846	C2/m	$\text{ZrBr} \rightarrow \text{Zr (P6}_3/\text{mmc)} + \text{Br (Cmce)}$	stable	-0.249	0.030
ZrGa_3^*	mp-30685	I4/mmm	$\text{ZrGa}_3 \rightarrow \text{Zr (P6}_3/\text{mmc)} + 3 \text{Ga (Cmce)}$	stable	1.225	0.000
ZrHg_3^*	mp-1608	Pm-3m	$\text{ZrHg}_3 \rightarrow \text{Zr (P6}_3/\text{mmc)} + 3 \text{Hg (P6/mmm)}$	stable	1.108	0.000
ZrMo_2^*	mp-2049	Fd-3m	$\text{ZrMo}_2 \rightarrow \text{Zr (P6}_3/\text{mmc)} + 2 \text{Mo (Im-3m)}$	stable	-6.870	0.003
ZrO_2	mp-775909	P4_2/mnm	$\text{ZrO}_2 \rightarrow \text{Zr (P6}_3/\text{mmc)} + \text{O}_2 \text{ (C2/m)}$	stable	-0.552	0.019
ZrRh^*	mp-2808	Pm-3m	$\text{ZrRh} \rightarrow \text{Zr (P6}_3/\text{mmc)} + \text{Rh (Fm-3m)}$	stable	-1.156	0.016
ZrTi	mp-1215236	P4/mmm	$\text{ZrTi} \rightarrow \text{Zr (P6}_3/\text{mmc)} + \text{Ti (P6}_3/\text{mmc)}$	not viable	-0.253	0.000

S8.2 Endobondic Reactions (Extension Batch)

Table S16: Decomposition-into-elements (case 1) reactions labeled **en-dobondic** ($\Delta\text{ICOHP} \geq 0$, products – reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)
0.3333 Be ₃ N ₂ → Be + 0.3333 N ₂	mp-1017550	0.2081	0.3960
BeCl ₂ * → Be + Cl ₂	mp-23267	0.0087	1.8204
BeF ₂ * → Be + F ₂	mp-15951	0.0004	3.1514
BeO* → Be + 0.5 O ₂	mp-7599	0.0153	2.7864
BeO → Be + 0.5 O ₂	mp-1778	0.0070	4.4391
0.3333 C ₃ N ₄ * → C + 0.6667 N ₂	mp-571653	0.4981	1.1208
0.3333 C ₃ N ₄ → C + 0.6667 N ₂	mp-2852	0.5033	1.8456
0.5 Cs ₂ S* → Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl* → Cs + 0.5 Cl ₂	mp-573697	0.0071	0.8830
CsF* → Cs + 0.5 F ₂	mp-8455	0.1088	3.8464
CsO ₂ * → Cs + O ₂	mp-1441	0.0349	0.0666
CsO ₂ → Cs + O ₂	mp-1096936	0.0940	0.7743
KCl* → K + 0.5 Cl ₂	mp-23289	0.0776	0.7616
0.5 Li ₂ O* → Li + 0.25 O ₂	mp-1960	0.0000	2.3366
0.5 Li ₂ O → Li + 0.25 O ₂	mp-755894	0.0844	1.9481
0.5 Li ₂ S* → Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li ₂ S → Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li ₃ N* → Li + 0.1667 N ₂	mp-2341	0.0039	0.6165
LiCl* → Li + 0.5 Cl ₂	mp-22905	0.0243	2.8030
LiCl → Li + 0.5 Cl ₂	mp-1185319	0.0039	2.5005
LiF* → Li + 0.5 F ₂	mp-1138	0.0000	4.4673
LiF → Li + 0.5 F ₂	mp-1009009	0.2875	4.5131
LiP ₅ * → Li + 5 P	mp-2412	0.0077	0.9043
LiP ₅ → Li + 5 P	mp-32760	0.0963	0.8692
MgCl ₂ * → Mg + Cl ₂	mp-23210	0.0000	0.1568
MgF ₂ * → Mg + F ₂	mp-1072956	0.0025	0.0015
MgF ₂ → Mg + F ₂	mp-1062842	0.2636	0.0229
0.5 Rb ₂ O* → Rb + 0.25 O ₂	mp-1394	0.0032	0.7235
RbCl* → Rb + 0.5 Cl ₂	mp-23299	0.0298	1.7924
RbF* → Rb + 0.5 F ₂	mp-2064	0.0516	3.9244
RbF → Rb + 0.5 F ₂	mp-11718	0.0000	1.9497
SiO ₂ * → Si + O ₂	mp-9258	0.5770	3.3332
SiO ₂ → Si + O ₂	mp-10064	1.2779	2.1735
SiO ₂ → Si + O ₂	mp-556044	1.4574	0.0958
SiO ₂ → Si + O ₂	mp-638049	1.7448	0.3849

S8.3 Exobondic Reactions (Extension Batch)

Table S17: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta\text{ICOHP} < 0$, products – reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)
AgN* → Ag + 0.5 N ₂	mp-1091417	1.5201	-7.5875
AgN → Ag + 0.5 N ₂	mp-1009766	1.5495	-6.4957

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1424734	1.9196	-8.0907
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1428312	1.6519	-5.3559
$0.2 \text{ Ba}_5\text{P}_4^* \rightarrow \text{Ba} + 0.8 \text{ P}$	mp-17566	0.0000	-0.9590
$\text{BaCl}_2^* \rightarrow \text{Ba} + \text{Cl}_2$	mp-23199	0.0068	-0.2244
$\text{BaF}_3^* \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1080821	0.0000	-0.8680
$\text{BaF}_3 \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1183303	0.0612	-0.2493
$\text{BaN}_2^* \rightarrow \text{Ba} + \text{N}_2$	mp-1102371	2.0568	-8.6248
$\text{BaO}^* \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1008500	0.0189	-0.3221
$\text{BaO} \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1950989	0.0434	-1.6129
$\text{BaS}_3^* \rightarrow \text{Ba} + 3 \text{ S}$	mp-556296	0.0436	-0.6233
$0.3333 \text{ Be}_3\text{N}_2^* \rightarrow \text{Be} + 0.3333 \text{ N}_2$	mp-18337	0.0000	-0.0746
$0.3333 \text{ Be}_3\text{P}_2^* \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-567841	0.0000	-0.2111
$0.3333 \text{ Be}_3\text{P}_2 \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-1084771	0.1567	-0.1795
$\text{BeCl}_2 \rightarrow \text{Be} + \text{Cl}_2$	mp-1172909	3.1688	-5.7807
$\text{BeF}_2 \rightarrow \text{Be} + \text{F}_2$	mp-975644	2.5181	-0.6816
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-1182484	0.5848	-3.6399
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-563	1.5119	-0.4498
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1047	0.0167	-4.4881
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-844	0.0000	-4.4374
$0.3333 \text{ Ca}_3\text{N}_2 \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1013524	0.5112	-4.4898
$\text{CaCl}_2^* \rightarrow \text{Ca} + \text{Cl}_2$	mp-23214	0.0011	-0.3199
$\text{CaCl}_2 \rightarrow \text{Ca} + \text{Cl}_2$	mp-1120739	0.0422	-0.2208
$\text{CaF}_2^* \rightarrow \text{Ca} + \text{F}_2$	mp-10464	0.0346	-0.2227
$\text{CaF}_2 \rightarrow \text{Ca} + \text{F}_2$	mp-554355	0.2403	-0.1814
$\text{CaN}^* \rightarrow \text{Ca} + 0.5 \text{ N}_2$	mp-1058549	0.3982	-4.9183
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1079707	0.2098	-2.2127
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-2605	0.0000	-0.9525
$\text{CaO} \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1181903	2.7281	-2.9971
$\text{CsN}_3^* \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-510557	0.0000	-2.2903
$\text{CsN}_3 \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-1225892	0.0113	-1.6664
$\text{CsP}_7^* \rightarrow \text{Cs} + 7 \text{ P}$	mp-1201790	0.0000	-0.2978
$\text{CsP}_7 \rightarrow \text{Cs} + 7 \text{ P}$	mp-1213206	0.0001	-0.2667
$0.5 \text{ K}_2\text{O}^* \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-971	0.0000	-0.4357
$0.5 \text{ K}_2\text{O} \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-1223784	0.2000	-1.5638
$0.5 \text{ K}_2\text{S}^* \rightarrow \text{K} + 0.5 \text{ S}$	mp-1022	0.0000	-0.3359
$\text{KF}_3^* \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-1544557	0.0125	-1.1511
$\text{KF}_3 \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-2749823	0.0119	-1.1505
$\text{KN}_2^* \rightarrow \text{K} + \text{N}_2$	mp-1071868	1.1630	-3.4244
$\text{KN}_2 \rightarrow \text{K} + \text{N}_2$	mp-1180757	1.2293	-4.2865
$\text{KN}_3 \rightarrow \text{K} + 1.5 \text{ N}_2$	mp-636056	1.0370	-5.9991
$\text{KP}^* \rightarrow \text{K} + \text{P}$	mp-1058315	1.0080	-2.1600
$\text{KP} \rightarrow \text{K} + \text{P}$	mp-1057787	1.0092	-2.1366
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	mp-569626	0.1991	-0.0862
$\text{MgO}^* \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-549706	0.1180	-1.4179
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-1009127	0.7484	-1.0763
$\text{MgS}^* \rightarrow \text{Mg} + \text{S}$	mp-1315	0.0000	-0.3401
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	mp-13032	0.0343	-0.4163
$0.5 \text{ Mn}_2\text{O}_7^* \rightarrow \text{Mn} + 1.75 \text{ O}_2$	mp-28338	0.3346	-1.4687
$\text{MnO}_2^* \rightarrow \text{Mn} + \text{O}_2$	mp-510408	0.0464	-0.5274
$0.5 \text{ Na}_2\text{Cl} \rightarrow \text{Na} + 0.25 \text{ Cl}_2$	mp-1077015	0.2391	-0.3965
$0.5 \text{ Na}_2\text{O}^* \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-2352	0.0000	-1.3057
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-1975297	0.0981	-1.5619
$\text{NaN}_3^* \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-22003	0.0017	-1.5201
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-1065265	1.0162	-5.4248
$\text{NaS}^* \rightarrow \text{Na} + \text{S}$	mp-409	0.0023	-0.4911
$\text{NaS} \rightarrow \text{Na} + \text{S}$	mp-1180106	1.0061	-2.3999
$\text{NbO}^* \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-2692	0.7888	-2.3179

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{NbO} \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-634965	0.8240	-3.1495
$\text{NbP}^* \rightarrow \text{Nb} + \text{P}$	mp-9830	0.6021	-2.4724
$\text{NbP} \rightarrow \text{Nb} + \text{P}$	mp-1220317	0.6209	-1.1731
$\text{OsC}^* \rightarrow \text{Os} + \text{C}$	mp-7142	0.9161	-2.0168
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009537	1.4795	-2.3184
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009539	1.3918	-3.1141
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-999308	0.9278	-2.9572
$\text{PbN}_6^* \rightarrow \text{Pb} + 3 \text{ N}_2$	mp-667338	0.1093	-2.5257
$0.5 \text{ Rb}_2\text{O} \rightarrow \text{Rb} + 0.25 \text{ O}_2$	mp-2023482	0.0479	-0.7592
$\text{RbN}_3^* \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-581833	0.0193	-1.3478
$\text{RbN}_3 \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-1219567	0.7966	-3.5636
$\text{RbP}^* \rightarrow \text{Rb} + \text{P}$	mp-1179688	0.9879	-1.9315
$\text{RbP} \rightarrow \text{Rb} + \text{P}$	mp-1058499	0.9882	-1.9104
$\text{RbS}^* \rightarrow \text{Rb} + \text{S}$	mp-558071	0.0019	-0.3217
$\text{RbS} \rightarrow \text{Rb} + \text{S}$	mp-1179684	0.9975	-2.7347
$\text{ReC}^* \rightarrow \text{Re} + \text{C}$	mp-1079494	0.6362	-2.6625
$\text{ReC} \rightarrow \text{Re} + \text{C}$	mp-1009731	1.1150	-1.9955
$\text{RuC}^* \rightarrow \text{Ru} + \text{C}$	mp-10030	0.6494	-2.1301
$\text{RuC} \rightarrow \text{Ru} + \text{C}$	mp-1009787	0.9097	-2.8605
$0.5 \text{ S}_2\text{N}^* \rightarrow \text{S} + 0.25 \text{ N}_2$	COD:4031496	–	-0.9767
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	COD:7017102	–	-1.3963
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-236	0.5087	-2.3179
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1078816	0.8090	-3.7082
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1173453	0.7225	-3.1673
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-684642	0.5498	-1.7362
$0.3333 \text{ Sr}_3\text{P}_4^* \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-14288	0.0000	-1.0472
$0.3333 \text{ Sr}_3\text{P}_4 \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-682953	0.0081	-1.0793
$\text{SrF}_2^* \rightarrow \text{Sr} + \text{F}_2$	mp-1102240	0.0646	-0.5421
$\text{SrF}_2 \rightarrow \text{Sr} + \text{F}_2$	mp-1102911	0.0267	-0.1481
$\text{SrN}^* \rightarrow \text{Sr} + 0.5 \text{ N}_2$	mp-1058586	0.4343	-4.5395
$\text{SrO}_2^* \rightarrow \text{Sr} + \text{O}_2$	mp-726705	0.0096	-1.9205
$\text{SrO}_2 \rightarrow \text{Sr} + \text{O}_2$	mp-2763179	0.0110	-1.7522
$\text{SrS}^* \rightarrow \text{Sr} + \text{S}$	mp-10627	0.2996	-0.1088
$\text{Ta}_2\text{N}^* \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1497	0.4516	-5.9309
$\text{Ta}_2\text{N} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009831	0.8163	-4.0540
$\text{Ta}_2\text{N} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009833	0.4917	-5.3385
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-1439	0.0045	-0.6461
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-9173	0.0575	-0.6509
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-430	0.0395	-0.6267
$\text{TlC}^* \rightarrow \text{Tl} + \text{C}$	mp-567465	2.1080	-3.1404
$\text{TlC} \rightarrow \text{Tl} + \text{C}$	mp-1058887	2.3757	-3.3638
$\text{TlN}^* \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1017982	0.7334	-3.7498
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1002222	0.9712	-2.3228
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1007778	0.7337	-3.1948
$\text{WC}^* \rightarrow \text{W} + \text{C}$	mp-13136	0.4477	-6.2684
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008630	0.8990	-5.9400
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008635	0.6729	-7.7315

S9 Additional Correlation Figures

Figures S1 and S2 reproduce, for reference, the bond-type-resolved correlation plots for the two baseline descriptors discussed in the main text (Section 3.2): the fully integrated reaction $\Delta(\text{ICOHP})$ and the raw (non-reaction) near-frontier antibonding population.

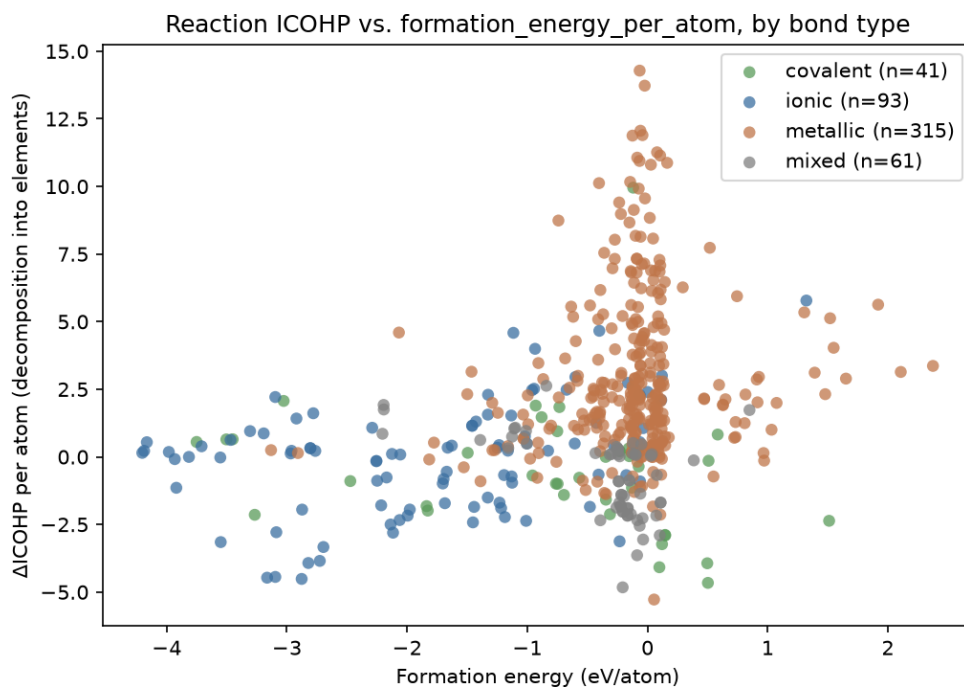


Figure S1: Fully integrated reaction $\Delta(\text{ICOHP})$ versus `formation_energy_per_atom`, by bonding-type subgroup.

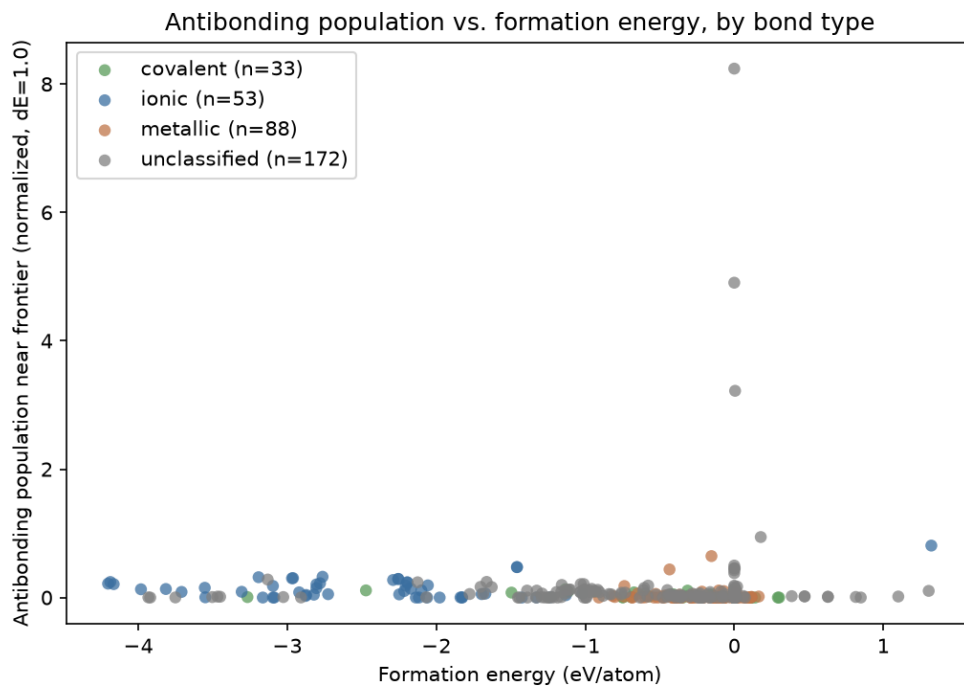


Figure S2: Raw (non-reaction) near-frontier antibonding population (ICOHP-based) versus `formation_energy_per_atom`, by bonding-type subgroup.